

Moonbase AndrewMills_1



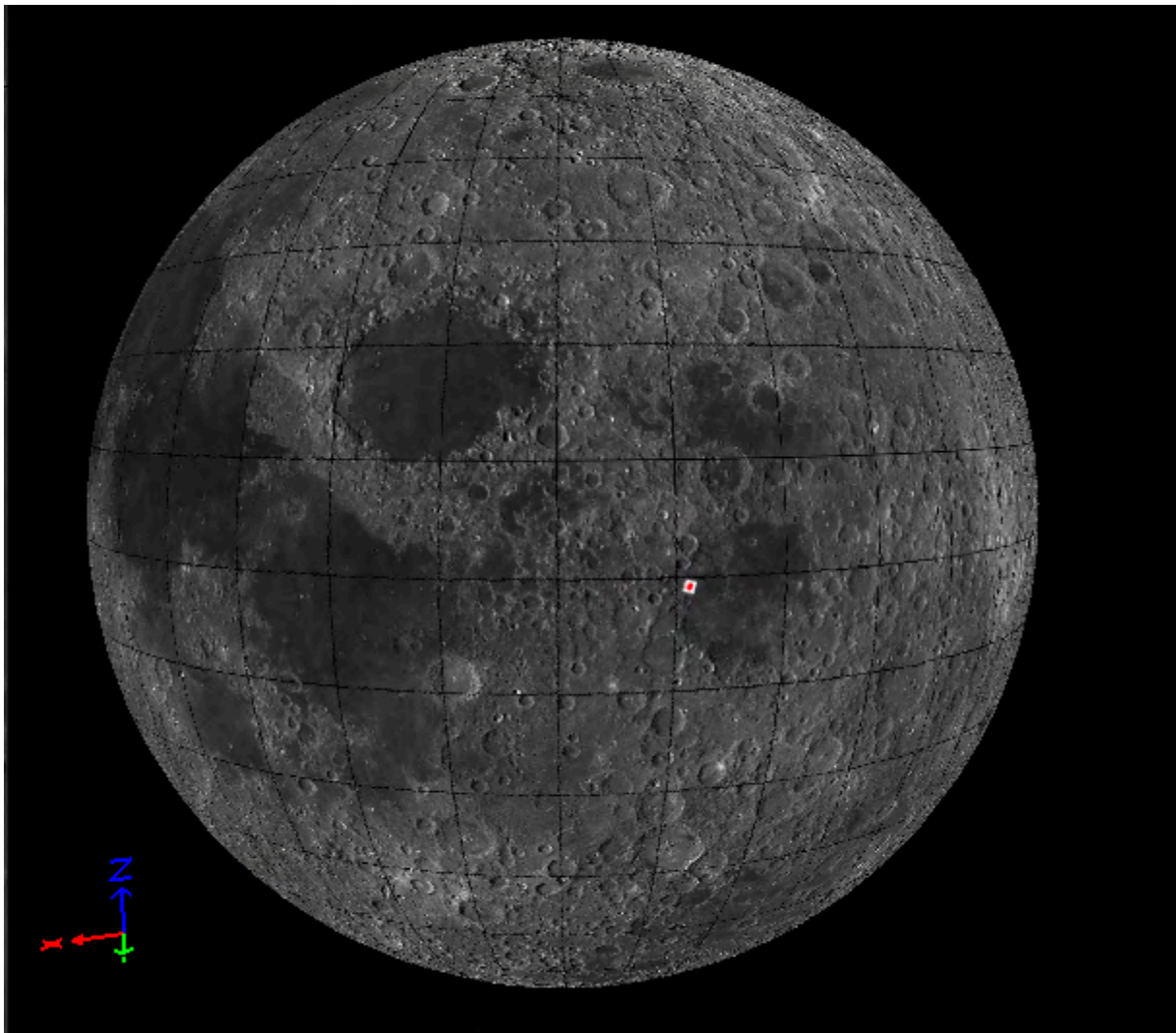
Report by - Andrew Mills

1.0 EXECUTIVE SUMMARY

This is a top-level description of the basic characteristics of Moonbase AndrewMills-1:

- This mission was designed to support a crew of 6 persons for 6 months (180 days).
- The base shall be located within Mare Smythii at coordinates 0.932 deg S, 81.217 deg E (shown in figure 1).

Figure 1
Location of Moonbase AndrewMills-1 (red dot)



- The purpose of this base is:
 - Base operations demonstration including surface and agricultural exploration
 - ISRU demonstration
 - Highlands/Mare selenology field study
- The base consists of:
 - Living Quarters
 - Astronaut support modules
 - Power Supply
 - Communications architecture
 - Launch/landing pad
 - Regolith processing modules
 - Laboratories
 - EVA garage
 - Consumable recyclers
 - Airlocks
- Supply needs:

The base will be assembled robotically prior to astronaut arrival. Consumables such as water, O₂, and food will also be supplied prior to the astronauts arriving. Fresh food will be grown, so Moonbase AndrewMills-1 will not require any resupply during a nominal 6 month mission.

2.0 THE MOON AND THE LUNAR ENVIRONMENT

The Moon is a stark, grey, pockmarked landscape with a difficult lighting environment, huge temperature swings, and a lack of any usable atmosphere. The Earth constantly hangs in the sky on the close side of the Moon, tidally locked along its entire 27 day orbit. Its regolith is jagged, with a retroreflective top layer similar to highway reflectors, nicknamed a “fairy castle” effect. Distances are hard to estimate due to a lack of scale, and

looking down-sun creates an effect similar to snow whiteout, where a step forward could land on an unseen obstacle or dip. The mares are flat and dark with gently rolling plains, while the highlands are lighter but rugged and cratered. The low gravity gently pulls at 1/6 the acceleration of Earth's, creating an environment where work is easier than weightlessness and walking looks more like skipping.

The local environment's effects on Moon base AndrewMills-1 include:

- Dust: Testimonial from Apollo missions shows that dust was an un-looked for but serious issue. It got everywhere, was hard to clean, and wore down moving parts at a quick rate.
- Radiation: GCRs, SPEs, and Solar Wind bombard the lunar surface unattenuated by any atmosphere, and without shielding, astronauts will experience high doses of radiation. In the case of SPEs, that radiation could become fatal.
- High temperature flux: From 120.00 Kelvin by night to a maximum of 373.89 Kelvin by day at the equator, temperature flux is a serious consideration and threat to systems in need of longevity.
- Slow ejecta-fall: Not a major concern, but may affect sensitive instruments.
- Meteorites: Low risk of impact to an astronaut, but a hab should be shielded. Flux of meteorites is lower on the Moon's surface than in low earth orbit.
- Seismic events: Although deep moonquakes occur, they aren't a big concern for a habitat.

3.0 SCIENCE AT THE LUNAR BASE

The Moon base will, aside from exploration demonstrations, carry out the following science:

- Space physics observations
- Astronomy
- Selenological study of mare/highlands

- Regolith-based crop growth

Space physics observations will require a Sun imager, Earth imager, Earth sounder, Solar constant instrument, and a Solar magnetic field instrument. This observatory is meant to study how the Earth-Moon system fits into the solar environment, and the effects of that environment over a 180 day mission period.

Astronomy will require a Lunar transit telescope, an Optical interferometer, a 4m telescope, a submillimeter interferometer, and a VLF interferometer. Use of the moon for an astronomy observatory affords observation opportunities unavailable with Earth-based telescopes or LEO observatories. Along with atmosphere-free observation, solar eclipse observation and VLF observation of Earth's aural radiation will be possible.

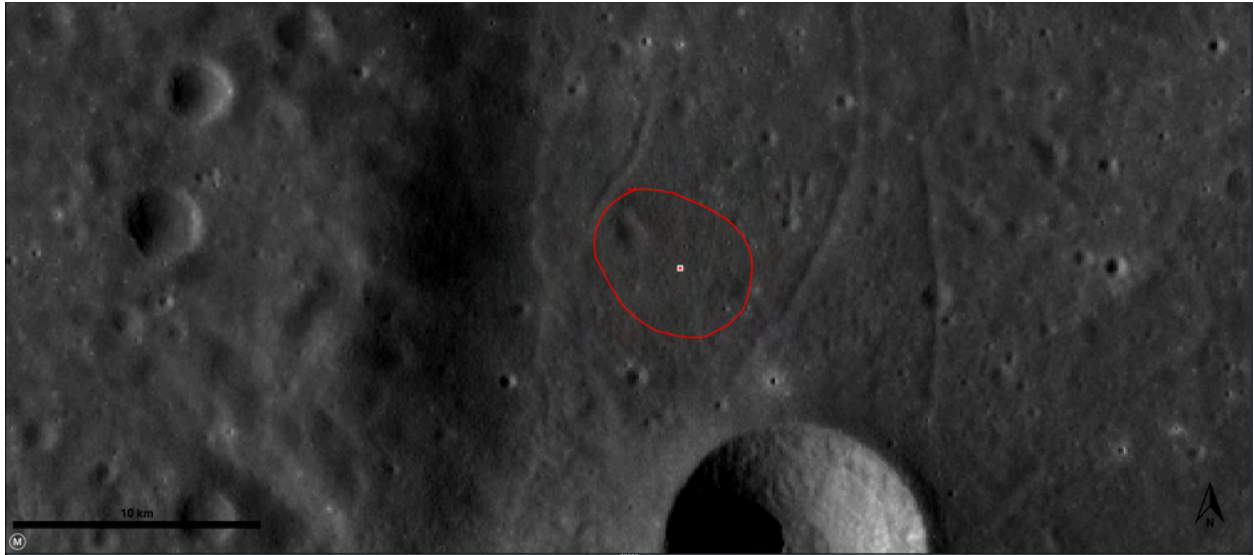
Selenological studies of the surrounding mare & highland regions would require robust transportation hardware for travel between the base in Mare Smythii to the nearby highlands, sampling tools, and lab equipment for onsite analysis. There is a need for transportation architecture that can traverse distances over 50 km in order to access certain highland regions. The selected region is not geographically isolated and has access to desirable diversity in selenological features nearby to the base.

Regolith-based crop growth studies will determine the possibility of adapting lunar regolith into a suitable growth environment for crops. Pioneer crops and supplementary nutrients will be used to help prepare the regolith.

4.0 LUNAR BASE SITING

Moon base AndrewMills-1 is located along the Moon's equator at coordinates 0.932 deg S, and 81.217 deg E, and is in the precise location presented in Figure 2.

Figure 2
10 km scale view of Moon base AndrewMills-1 using LROC data



According to the findings of the “Site Selection Strategy for a Lunar Outpost” workshop, a Moon base would benefit from a location with the following characteristics:

- Within 10 degrees of the lunar equator to maximize access to the celestial sphere, and within 2 degrees of the lunar equator for space physics observations of the Sun.
- Within 5 to 10 deg of the mean lunar limb to maintain Earth in constant FOV low on the horizon.
- At a mare-highland contact on a mare surface for selenological/ selenophysical exploration.
- A mature regolith is required for solar wind implanted resources; mare material offers the most flexibility in resource extraction.
- Some space physics experiments require regions of low KREEP (Potassium, Rare Earth Elements, Phosphorous).
- The outpost should have a “flat” area for emplacement of instruments. Hills or large craters could offer design advantages for interferometers. The selected location should allow placement of some instruments at several kilometers distance to minimize operation interferences.

The Avery crater location meets these conditions. It is within the latitude requirements of observatory instruments (-0.932 deg N) and the longitude requirements of communications (81.217 deg E). It is located in Mare Smythii within excursion distance (13.6 km) of a highlands region. KREEP has a low concentration in Mare Smythii, so space physics experiments will be unimpeded. There are flat, open spaces for instruments and the habitat, as well as craters and hills within excursion distance.

5.0 MOON BASE ELEMENTS AND SURFACE INFRASTRUCTURE

A mixed configuration of radial and courtyard was chosen for Moonbase AndrewMills-1 (shown in figure 3). The radial design allows for quick access to each of the base's various modules, while the courtyard approach allows for multiple ways to access each module in case of emergency. The base has three distinct wings meant to separate the "clean" and "dirty" functions. All of the major habitation modules are located among the clean wings, while all of the lunar regolith-facing modules are located among the dirty wing. "Dustibules" will be located at the intersection between the clean and dirty modules, where PPE will be donned and doffed. Modules included in the base are:

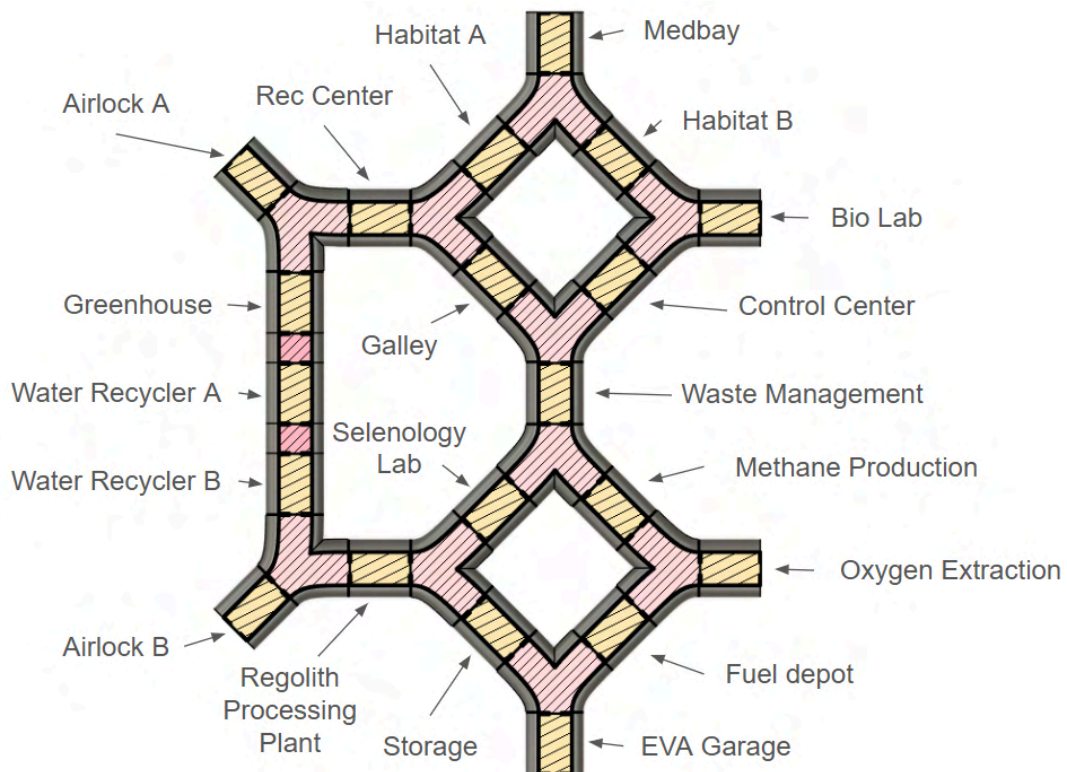
- 2 Habitats
- 1 Galley
- 1 Control Center
- 1 Recreation center
- 1 Greenhouse
- 1 Medbay
- 2 Water recyclers
- 1 Waste management module
- 1 Biology lab
- 1 Selenology lab
- 1 Methane production module
- 1 O₂ extraction module
- 1 Regolith processing plant
- 1 Storage module

- 1 EVA garage
- 2 O₂ storage tanks
- 2 Nitrogen storage tanks
- 2 Water storage tanks

Separate modules include:

- Launch/Landing pad
- 3 Communications tower modules
- 1 Observatory module
- 1 Solar array
- 1 Battery storage
- 1 Nuclear reactor
- 4 Navigation beacons (not pictured in Figure 3)

Figure 3
Layout of Moonbase AndrewMills-1



The two crew habitats are located along the edge of the courtyard of the north wing, connected to the medical bay, supply tanks, and recreation center for quick access. The control center and biology lab will also be located in the north wing, as they are clean workspaces. The greenhouse will be located to the north side of the west wing, adjacent to the water recycler modules and airlocks. The waste management module connects the north and south wings, and will act as an area for cleaning off traces of lunar dust. At the southern tip of the base will be the EVA garage, adjacent to the fuel depot for rover charging & to the storage module for regolith unloading. The regolith will move through the storage module to the regolith processing plant, selenology lab, methane production module, or the O₂ extraction module.

6.0 LUNAR BASE HABITAT LAYOUTS

Modules will be cylindrical, with dimensions of 5m length and 5m diameter. Module interiors will all fit a mold of 5m long by 2.75m wide floor space, with 3.5m of ceiling clearance. Extra space not filled by this rectangular prism of usable space will be filled with water, power, and air distribution. Lighting fixtures and air vents will be placed on the ceiling of each module. Individual floor plans for each module are shown in appendix 4.0. Free floor space and free volume are shown in table 1.

Table 1
Free space available within modules

Module	Free floor area (m2)	Free volume (m3)
Habitat (2)	6.88	24.08
Lab (2)	8.82	30.87
Production (6)	7.23	25.305
Medbay	7.62	26.67
Rec Center	10.62	37.17
C&C	8.88	31.08
Galley	8.61	30.135
Storage (3)	5	17.5
I Joint	3.255	11.3925
Y joint	28.343	99.2005
L Joint	11.049	38.6715
Total	384.117	1344.4095

Moonbase AndrewMills-1 will have two airlocks. They will each be located at the top and bottom of the west wing. Both will have the same design. Hatches will be 1.016m by 1.524m to allow EVA suits to pass through with minimal sealing length along the door.

All modules conform to the NASA Langley research center's "Foundation Surface Habitat Function-Driven Net Habitable Volume," (December 2020) for comfortable station-living. Individual floor plans for each module, as well as for each airlock, are available in appendix A4.0

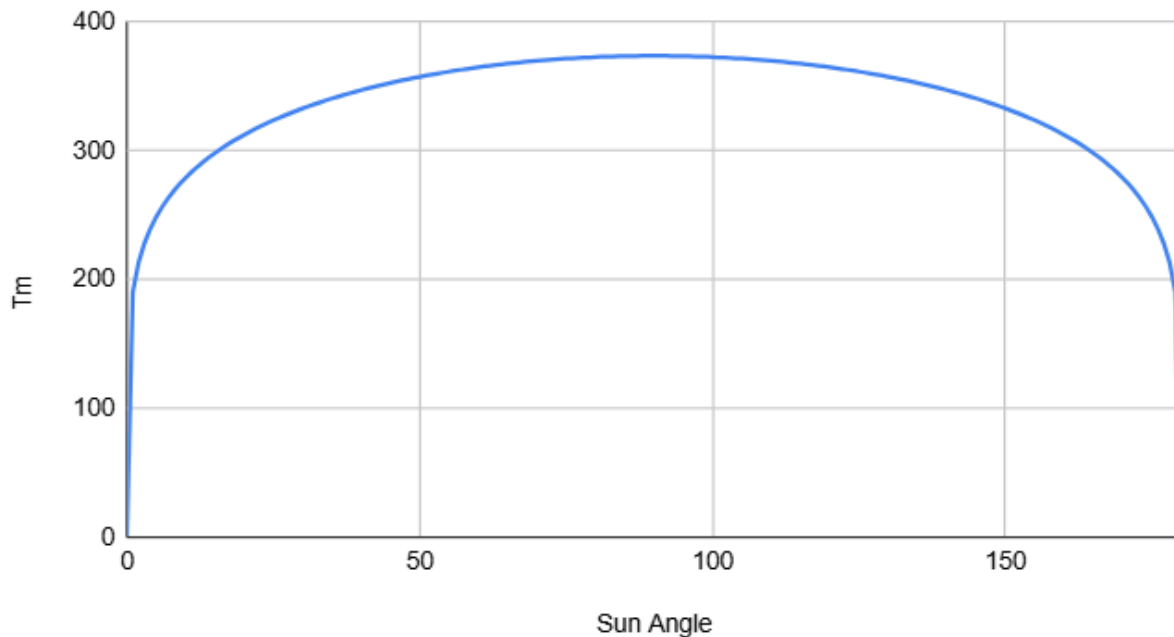
7.0 THERMAL CONTROL SYSTEMS

7.1 Surface temperature during a lunar day/night at Mare Smythii:

- During lunar day, the surface temperature varies from a min of 120 Kelvin to a max of 373.89 Kelvin (Figure 4).

Figure 4
Surface temperature vs. sun angle over a lunar day

T_m vs. Sun Angle



- During lunar night, a constant temperature of 120 Kelvin surface temperature is maintained.

7.2 Ideal Module Temperature

The ideal temperature depends on the module, but all should maintain a comfortable shirtsleeve environment for astronauts. In the habitat, lab, office, recreation, storage, and medbay modules, 20 degrees Celcius will be passively maintained. Production modules will need more thermal control (more cold plates) as their components will generate significantly more heat. Estimates for daily heat output of major base systems (based on Earth analogs) are:

- Astronauts: 100 W/hr each
- Personal computers: 350 W/hr each

- Health monitors: 100 W/hr each
- LED lighting: 30 W/hr each
- Production: 500 W/hr each machine (Could be higher)

7.3 Heat Input and Output

The greatest amount of heat transfer will come from radiation, as the sun's radiation will create immense heat, while the space background will expose the surface to deep cold. Some heat will be added to the system by reflection from sunlit lunar surfaces. The base will conduct heat through the lunar regolith.

7.4 Active Thermal Control

Cold plates will be installed in each module where each electrical/production component will interface for heat rejection. An internal thermal control system (heat collection) will run heat pipes to each cold plate, collecting heat and transferring it to the external thermal control system (heat rejection). This system will use radiators to reject heat through radiation to deep space. Radiators function by passing a hot working fluid through a high surface-area metal plate, which radiates heat through a high emissivity coating into deep space.

7.5 Passive Thermal Control

The goal of the passive thermal control system will be to raise the heat capacity of the moon base, allowing it to maintain temperature across the temperature swings of the lunar day and night. The moon base will use a layer of regolith to bury each component, shielding the base from cosmic radiation, and raising its heat capacity. Lunar regolith has a high heat capacity, so this method of passive thermal control will be highly effective at maintaining a semi-constant temperature.

8.0 ENVIRONMENTAL HAZARDS

8.1 Radiation Hazards

Radiation can cause serious and in some cases fatal damage to astronauts, as well as significant damage to hardware and electrical systems. Solar Particle Events (SPEs) are rare, intense bursts of medium energy protons. Galactic Cosmic Rays (GCRs) are a constant stream of high energy ions. Crew radiation dose is a significant factor in the length of safe missions and crew health. Each type of radiation does damage on a different scale represented by a quality factor Q shown in table 2 (ICRP, 1977). The absorbed dose times Q gives the dose equivalent in Sievert (Sv), a useful tool for predicting the effects of absorbed dose. The radiation flux on the moon is high enough to require shielding for missions longer than sorties.

Table 2
Quality factors and Occurrence of Different Kinds of Radiation

Radiation	Q	Occurrence
X-rays	1	Radiation belts, Solar radiation, Bremsstrahlung
5 MeV γ-rays	0.5	"
1 MeV γ-rays	0.7	"
200 keV γ-rays	1.0	"
Electrons	1.0	Radiation belts
Protons	2–10	Cosmic radiation, Inner radiation belt
Neutrons	2–10	Close to Earth, the Sun and any matter
α-particles	10–20	GCR
HZE-particles		"

8.2 Temperature Hazards

High lunar day temperatures expose unprotected astronauts to heat exhaustion, while low lunar night temperatures expose astronauts to the risk of frostbite and hypothermia. These thermal issues necessitate robust thermal control systems.

8.3 Meteorite Hazards

Meteoroids travel at extremely high speeds, causing a potential penetration hazard for space operations. A model has been developed to predict the average flux of meteoroids on the lunar surface, showing 1088.42 impacts/m²/year of 1e-4 cm or greater impacts to 1.61142e-18 impacts/m²/year of 700 cm or greater impacts. This flux means that meteoroid shielding would be required to protect astronauts from the risk of hypoxia and DCS that could come from a meteoroid impact to the base.

9.0 SHIELDING REQUIREMENTS AND CONCEPTS

Regolith shielding provides a low-weight, passive, reliable solution to the shielding requirements of:

- Radiation effects: brought down to acceptable levels with 1-2 m of regolith shielding.
- Thermal effects: brought down to acceptable levels with 0.1-10 m of regolith shielding.
- Meteoroid effects: brought down to acceptable levels with 0.5 m of regolith shielding.

Moonbase AndrewMills-1 will use 2 m of regolith shielding to meet the requirements of each lunar hazard without needing to dig a trench or into a lava tube. There are four possible configurations of lunar regolith shielding for Moonbase AndrewMills-1:

- Case 1: Regolith retainer structure
- Case 2: Pyramid structure
- Case 3: Flat bin structure
- Case 4: Semicircle bin structure

Calculating the required regolith for each case (table 3), Case 4 would be best for my base since it uses the least regolith per module, for a total of 3294 m³ of regolith for the entire base.

Table 3
Regolith volume per module (m³)

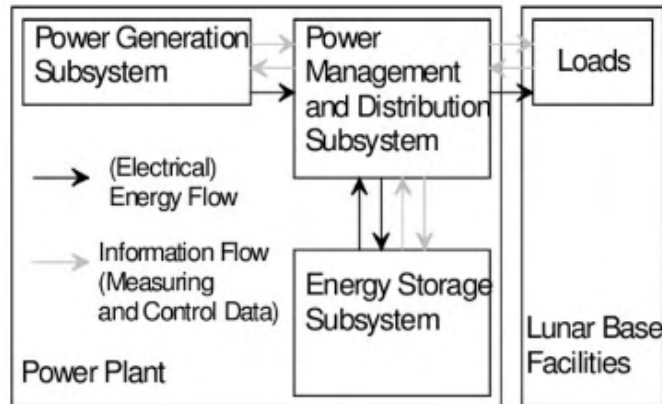
Empty module volume	98.17477042
Case 1	219.9114858
Case 2	214.245
Case 3	216.8252296
Case 4	173.3683577
Modules	19
Minimum regolith volume	3293.998796

The regolith would be deposited prior to astronaut arrival using autonomous robotic systems, which will be reused during astronaut operations for regolith collection.

10.0 POWER SUPPLY AND ENERGY STORAGE

Power generation, power management, energy storage, and loads are the basics of any power system, working together as shown in figure 5. Moonbase AndrewMills-01 will use a DOE solar cell array and Kilopower Reactor Using Stirling Technology (KRUSTY) nuclear reactors for power generation. Solar and nuclear will both be used to ensure continuity of power generation during the lunar day and night cycle. Lithium-Sulfur batteries will be used for energy storage, and a network of cables will distribute power to loads throughout the Moon base.

Figure 5
Figure 5 - Breakdown of power handling



The solar array will be located at least 1 km to the southwest of the habitable section of Moonbase AndrewMills-01 to avoid the dust plume from the launch/landing pad. The KRUSTY reactors will be placed close to the solar farm to share power distribution cables. The energy storage bay will be located close to the south wing of the base for easy maintenance access.

Colozza 2020 suggested that the power demands for a six-personnel lunar base would approach 25.8 kW for ISRU, and 28.05 kW for life support/science use, totalling 53.85 kW. With a mostly passive thermal control system, there would be no difference in day and night power usage.

10.1 Photovoltaic Power

Solar panels work by converting sunlight into electricity using photovoltaic cells. These cells are made up of semiconducting materials. The current most commonly used solar cell is made of Gallium Arsenide, or GaAs. According to the Department of Energy, a new, highly efficient type of solar cell is being tested that involves using multiple junctions to capture a larger slice of the wavelengths included in sunlight, with a theoretical peak efficiency of over 45%. Using these solar cells instead of GaAs cells could lead to a significant reduction in required solar array area. Using

GaAs solar cells, 123.736 m² of solar panels would be required to provide the full power to the base. However, Moonbase AndrewMills-01 will be using DOE solar cells, requiring a 87.990 m² solar array.

10.2 Nuclear Fission Power

KRUSTY reactors are stirling generators, utilizing fission as a heat source. They have been successfully tested on Earth, and for the purposes of this base, are assumed to be capable of generating 10kW of power each. With an estimated power requirement of 53.85 kW, 6 KRUSTY reactors would be required to meet that need.

10.3 Energy Storage System

Lithium-Sulfur battery systems are a significantly higher specific energy solution to power storage compared to Lithium-ion batteries, so despite Li-ion's many benefits, Li-S will be used. Battery storage will involve a network of secondary batteries receiving power from the solar array and KRUSTY reactors, and providing power to base systems as needed. This will allow for solar power generated during lunar day to be used during lunar night, or for surplus power to be accessed during peak usage times.

The total energy generated in a lunar day (14 Earth days) by both the solar array and KRUSTY reactors would be 38.2536 MWh. Li-S batteries have a gravimetric energy density of 2510 Wh/kg, so 1524.05 kg of batteries will be required.

11.0 LUNAR BASE LIFE SUPPORT AND CREW HEALTH

11.1 The Need for a Life Support System

With 6 crew spending 6 months on the lunar surface, the need for robust life support systems is evident. Human beings need input and create output, but with some processing, most outputs can be recycled into usable

consumables. Life support systems keep the harsh lunar environment livable while also recycling yesterday's coffee into tomorrow's.

11.2 Facets of Life Support

Table 3 shows the overarching work areas of work for the life support system, and a more comprehensive list is provided here:

- Provision of O₂
- CO₂ Removal
- CO₂ Reduction
- Atmosphere composition monitoring
- Temperature control
- Pressure control
- Humidity control
- Particulate control
- Trace contaminant control
- Fire detection and suppression
- Provision of potable water
- Provision of hygiene water
- Water quality monitoring
- Urine and feces removal
- Urine processing
- Waste removal
- Waste processing
- Provision of food
- Refrigeration

Table 3
Main Areas of Life Support

Life Support Area	Purpose
Atmosphere Management	Atmosphere composition control, temperature and humidity control, pressure control, atmosphere regeneration, contamination control, ventilation
Water Management	Provision of potable and hygiene water, recovery and processing of waste water
Food Production and Storage	Provision and storage of food
Waste Management	Collection, storage, and processing of human waste and trash
Medical and Safety Aspects	Medical care, fire detection suppression, radiation shielding

11.3 How Moonbase AndrewMills-1 Accomplishes Life Support

- Atmosphere management: air will be constantly flowed through all habitats using fans, in through a vent in the ceiling and out through vents near the floor. Recollected air will be scrubbed of CO₂ and contaminants with a similar system to the ISS before being re-released.
- Water management: Two water-recycling modules in the east wing provide the means to clean used water, where a system similar to that on the ISS will provide clean water back to the astronauts.
- Food production and storage: Shelves in the galley and a storage module in the south wing allow storage of food, while preparation will take place in the galley.
- Waste management: Trash and human waste will be collected, and whatever can be recycled will be. Urine can be cleaned and reused, and trash may serve other purposes, such as food waste and feces as compost.
- Medical and safety aspects: The medbay provides astronauts with medical means. Fire detectors will be embedded in the air-scrubbing system, and module hatches will be sealable to slow fire spread. Radiation shielding is available with the regolith shield.

11.4 Flow Chart of the Life Support System

Using the data from figure 6 in table 4, for 6 crew, 25.65 kg of consumables are required each day, totaling 4617 kg for the mission duration. Since Moonbase AndrewMills-1 will use a 98% effective water-reclaiming system and 50% effective CO₂ recycling system, their daily needs will be recycled. As shown in table 5, with these recycling systems, only 1112.4 kg of consumables will be required for the mission duration.

Figure 6
Needs and waste for 4 crew

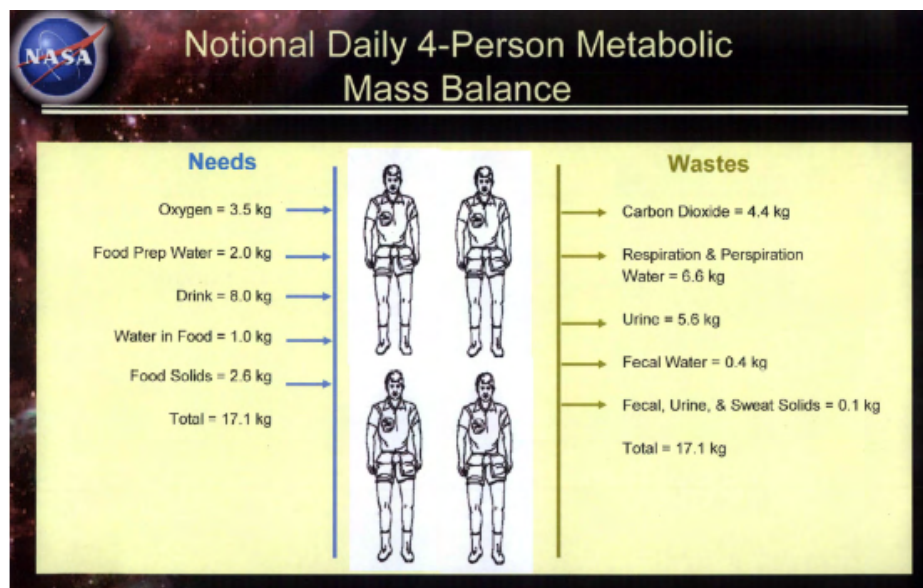


Table 4
Needs and wastes for 6 crew daily

6 Crew			
Input	Amount	Output	Amount
Oxygen	5.25	CO2	6.6
Food prep water	3	Respiration and perspiration water	9.9
Drink	12	Urine	8.4
Water in food	1.5	Fecal water	0.6
Food solids	3.9	Waste solids	0.15
Total	25.65	Total	25.65

Table 5
Consumable needs over mission duration with recycling

With ISS levels of reclamation		
Consumable	Daily	6 months
Oxygen	1.95	351
Water	0.33	59.4
Food	3.9	702
Total	6.18	1112.4

11.5 Dietary & Nutritional Requirements

Based on a required 2500 cal per day per person, options must be chosen for breakfast, lunch, dinner, snacks, sides, dessert, and drinks. These options not only must satisfy the basic needs of astronauts, but also must provide a cultural and personal mix of options. Most mixes of one each of these options will satisfy the daily minimum calories, add snacks if it isn't enough (1735 min to 3450 max).

Breakfast options should be hardy to provide for a long, strenuous workday. The options I have chosen are:

- Ribollita stew (400 cal): A Tuscan peasant food that was meant to keep farmers warm during cold harvests.
- Microgreen omelette (400 cal): A worldwide breakfast staple with fresh, Lunar-grown microgreens for nutrition.
- Empanadas with Chimichurri sauce (375 cal): An Argentinian hardy breakfast with a flavorful sauce on the side.

Lunch options should be quick to eat and contain fats for usable energy.

- Satay Ayam (425 cal): An Indonesian protein-rich lunch.
- Vegetable samosas (750 cal): An Indian dish that's easy to heat and eat.

- Bánh Mì (600 cal): A Vietnamese baguette sandwich good for eating with the team.

Dinner options should be the most calorie-dense of the day, and should contain carbohydrates for healthy energy storage.

- Chicken pad see-ew (600 cal): A Thai meal with plenty of protein and carbs.
- Honey sriracha chicken meatballs with jasmine rice (750 cal): A slightly spicy, delicious meal that carries a lot of calories.
- Vegetable lasagna (800 cal): Great meal for using the fresh vegetables from the greenhouse.

Snacks should be good for a quick energy boost and taste good, but should be healthy to avoid overeating.

- Beef jerky (80 cal): Hardy snack for
- Protein bar (300 cal): Quick energy boosting snack.
- Baby carrots (35 cal): A great hydrating snack for when you're busy or driving (a rover).

Sides are meant to be eaten with any meal to add volume in case the astronaut is still hungry, or if more calories are needed.

- Green salad (150 cal): A good use for fresh greenhouse vegetables.
- Biscuits (300 cal): A good way to add carbs and warmth to a meal.
- Mac and cheese (500 cal): A good starch to add to most meals.

Desserts are there to keep the crew happy and full.

- Ice cream (200 cal): Good for eating around a table with the crew. An astronaut needs ice cream.
- Chocolate chip squares (150 cal): Good at any temperature and can eat on the go. A favorite of mine growing up.

- Peach cobbler (550 cal): Best served hot, great for warming up after a cold day.

Drink options should have a caffeine boost and taste good, or provide one or the other.

- Coffee (2 cal)
- Tea (0 cal)
- Soda (150 cal)

11.6 Crops Grown at Moonbase AndrewMills-1

There will be three growth environments within the base. Here are the environments and the primary groups to be grown in each environment:

- Conventional soil: For the purpose of seeing how crops grow in a low-gravity environment, as well as feeding the astronauts. These will be the primary crops of the conventional soil environment:
 - Sweet potatoes
 - Tomatoes
 - Peanuts
- Hydroponics: For the purpose of providing the bulk of the astronaut food. Hydroponic troughs will line the joint modules for the sake of growth volume and for a pleasant walking experience around the base.
 - Lettuce
 - Mung beans (microgreens)
 - Green onions
- Moon regolith/soil mix: For the purpose of seeing if pioneer crops can implant nutrients into moon regolith during their growth process, while also potentially providing a food source to astronauts.
 - Soy beans
 - Beets
 - Nopales

11.7 Crew Selection Factors

The crew will be selected starting with the same selection prerequisites that NASA currently requires, while adding additional select-in factors:

- Emotional maturity & stability: The candidate must display a stable mindset and maintain emotional maturity.
- Healthy self-esteem: Candidates can't beat themselves up over mistakes, and instead should learn from them.
- Stable relationship-building: Must work well in teams, and build a good relationship with crew.
- Expressivity: Candidates should be able to express emotions or needs adequately and in a healthy way.
- Sense of humor: Everyone wants to work with someone who can keep the mood lighthearted, but knows when to buckle down.
- Insight and self-awareness: Candidates should know their limitations and should be able to use their expertise to creatively solve problems.
- Cultural sensitivity: The crew will have different cultural norms, and those must be understood and accepted by the candidate.

11.8 Example Crew Selection

- Crew commander/CMO/EVA: I would be the crew commander/Crew medical officer. I've led multiple teams as an aerospace engineer, am working on an EMT-Basic certification, and possess the relationship building, self-esteem, and self-awareness skills necessary to lead a long term mission.
- Mission controller/astronomer: My fiance, Ava, would make a great mission controller/astronomer. She is close to earning a Phd in astrophysics and displays excellent skills in emotional maturity, cultural sensitivity, and relationship building.
- Biologist/Doctor: My mom, Jody, would make an excellent biologist/doctor astronaut. She is a physician with an undergrad

background in biology, and shows the insight, expressivity, and sense of humor necessary to make a great part of the lunar crew.

- Selenologist/artist: My friend and neighbor Basem would be a great selenologist/artist astronaut. Working as an industrial designer, he would be an asset on a base where creative thinking is needed. He shows insight, tenacity, and a sense of humor.
- Engineer/IT: My grandma Joan would make a good crew engineer/IT astronaut. As one of the first computer programmers in the United States, she learned all she would need as an IT astronaut. She also shows the expressivity, sense of humor, and self awareness necessary for a good crew engineer.
- Engineer/EVA: My friend John would work well as an EVA engineer. He's working on a degree in space systems, and possesses the insight, self-esteem, and stability necessary for a cool-headed EVA astronaut.

11.9 Medical Care

Capability to treat medical events of class I and II must be present on Moonbase AndrewMills-1, and the ability to rapidly launch in case of the incidence of class III must exist:

- Class I: A minor medical event that would require some type of medication or medical personnel intervention without preventing the crew member from fulfilling their tasks.
- Class II: A medical event that could be supported at the lunar base, but would prevent the crew member from continuing their tasks.
- Class III: An unsupportable medical event that could lead to death or permanent disability of the crew member if not returned to Earth as soon as possible.

Crew mental health will also be a huge priority for maintaining crew effectiveness and safety. Weekly private counselling appointments, group meals, and greenery along walking paths would help astronauts feel comfortable in their working environment.

12.0 COMMUNICATION AND NAVIGATION SYSTEM

12.1 Communications and Navigation Requirements

The communication and navigation systems need to handle/provide:

- Data from engineering and science measurements
- Radiometric data for navigation
- Video data for operations
- Communication with Earth and between base components
- Privacy, security, reliability, and redundancy
- Position data to astronauts on EVA

12.2 Communications

Moonbase AndrewMills-1 will have a 10m communications tower located close to the west wing of the base, a repeater alongside the western highland ridge navigation beacon (figure 7), and another repeater at the eastern mare navigation beacon. The western highland ridge tower will have a 180 degree view of all unobstructed regions of Mare Smythii in a 73.61 km radius (radius shown in figure 8) since it will be located at the top of a 1.55 km elevation slope above this region. Shorter range LOS communications will be available from the base tower and eastern Mare tower in a radius of 5.89 km.

Figure 7
Locations of the western ridge tower, base tower, and eastern mare tower

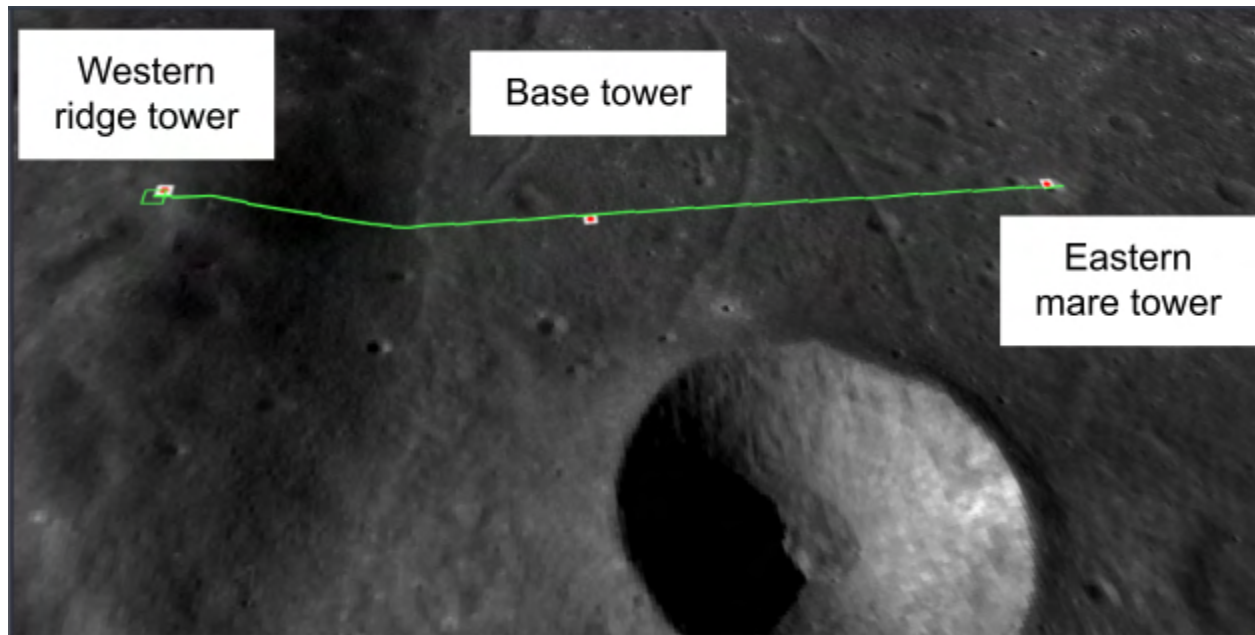
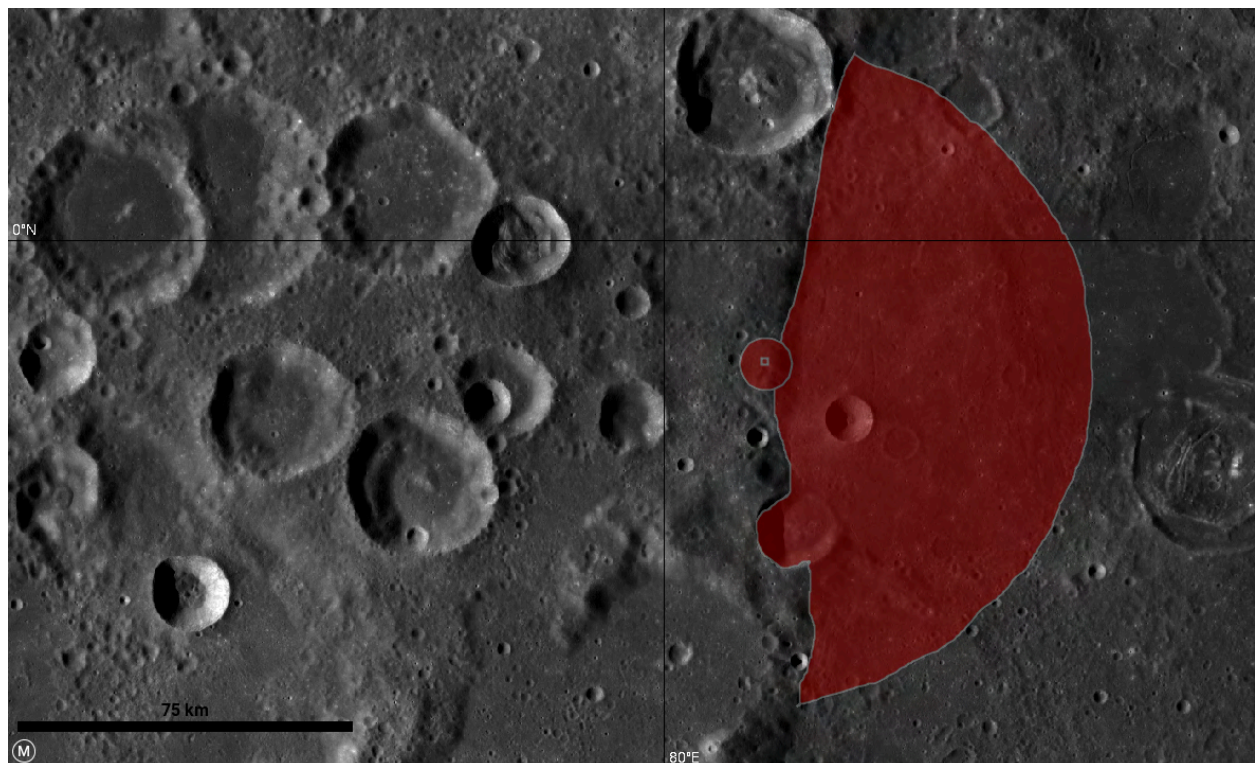


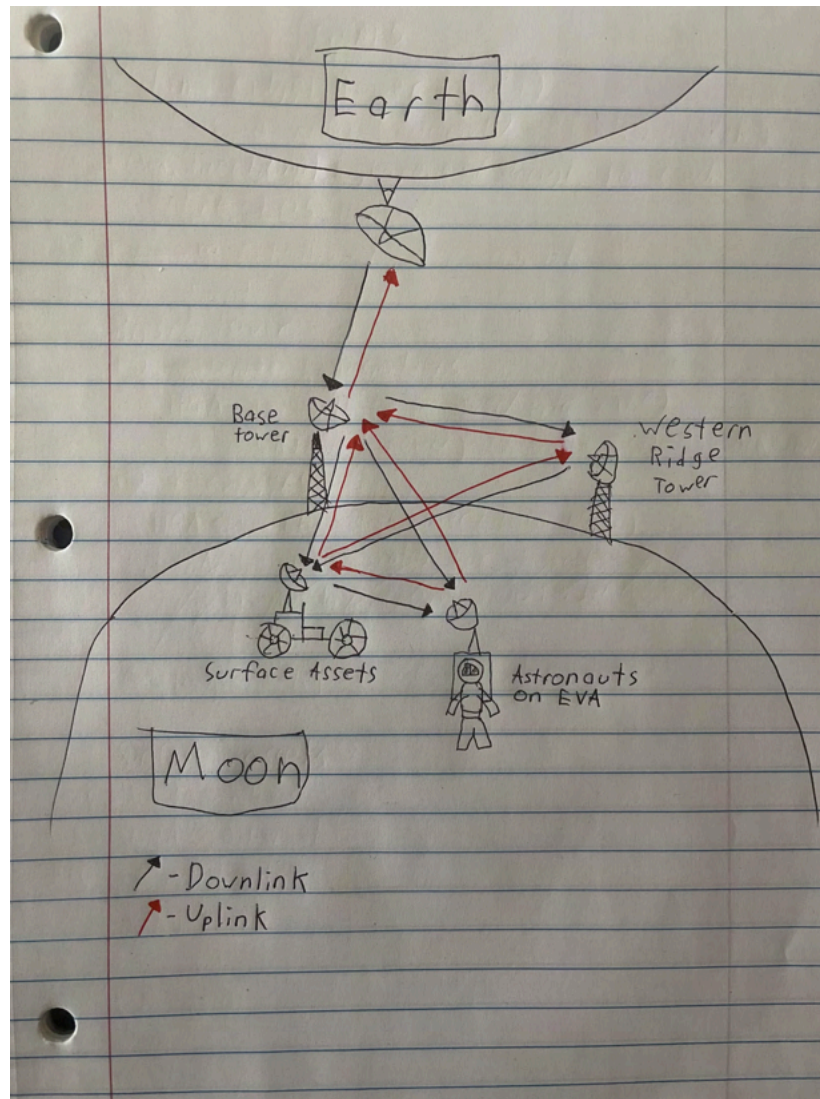
Figure 8
LOS range of the western ridge tower in red



Users of the communications system will have the following communications capabilities in the architecture shown in figure 9:

- Astronauts on EVA
 - Low gain transmitter
 - Low gain receiver
- Surface assets
 - High gain transmitter
 - Low gain transmitter
 - Low gain receiver
- Science experiments
 - High gain transmitter
 - High gain receiver
- Western ridge tower
 - High gain transmitter
 - Low gain transmitter
 - Low gain receiver
- Earth based ground support
 - High gain transmitter
 - High gain receiver

Figure 9
Communications architecture



12.3 Navigation

There won't be lunar GPS satellites since the base will constantly be within view of the Earth, and will be able to navigate using ground systems. Moonbase AndrewMills-1 will have three navigation beacons to provide fixed points to navigate from:

1. Base beacon: Located at the base of the communications tower.

2. West highland ridge beacon: Located on top of the ridge of the hill to the west of the base.
3. Eastern mare beacon: Located to the east of the base further into Mare Smythii and past the Moon's mean limb, so out of Earth LOS for some time.

13.0 LUNAR EVAS AND LUNAR SURFACE VEHICLES

There are three types of EVAs that the astronauts of Moonbase AndrewMills-1 will expect to perform:

1. Scheduled EVA: Planned before the mission and included in nominal mission timeline.
2. Unscheduled EVA: Not included in nominal timeline but may be necessary to achieve success.
3. Contingency EVA: Required for safety of the crew, such as evacuation or emergency return to the habitat.

EVA tasks may range anywhere from site preparation to field maintenance. Expected lunar base EVA tasks are laid out in table 6.

Table 6
Candidate EVA Tasks at Lunar Base

Site Preparation	Surveys Major rock removal Smoothing Dust control Navigation aid setup
Habitat Installation	Hole excavation Framing/tie down Module transport Unloading Deployment Antenna deployment

Shielding installation	Regolith supports Regolith backfilling Clearing access paths
Science	Sample collection Experiment installation Location mapping Observatory setup
Power/TCS Systems	Site prep Deployment and Assembly Radiator activation Distribution connection Nuclear shielding excavation
Logistics Management	Unpacking Transport Transfer Storage Waste disposal Storage of spent recyclables
Resource Operations	Resource process setup (Pressure vessels, ore holding bins, plumbing, gas holding, heat exchangers, power sources, TCS) Excavation for processes
Lander Operations	Servicing Refueling Pre-launch checkout Relocation to depot/maintenance
Mining Operations	Equipment transport System setup System relocation
Maintenance	Inspections Field checks/measurements Replacements Repairs

There will be two unpressurized LTVs following Lunar Outpost's design, and one pressurized LTV following JAXA's design at Moonbase AndrewMills-1. These numbers allow for redundancy and flexibility in LTV operations.

JAXA's pressurized "Lunar Cruiser" shown in figure 10 has been developed jointly with Toyota, and is capable of performing the following tasks in the full range of expected lunar temperatures:

- Long-range surface transportation: Required for access to the complex selenology of the western highland regions and for the exploration of Mare Smythii.
- Secondary habitat: In case of need for evacuation from Moonbase AndrewMills-1, the pressurized rover will be able to serve as a temporary life boat while astronauts can either EVA to repair the base, or evacuate to the launch pad.

Figure 10
JAXA Pressurized LTV



The unpressurized LTV option shown in figure 11 was chosen for its versatility. With a modular payload interface, the unpressurized LTV could fill any role required of it. It will be useful for site prep and equipment

transport with a capacity of 1600 kg in 3 cubic meters. It will be incapable of operation during the lunar night, but its batteries will keep it functional until night passes. The unpressurized rover will fill the following roles:

- Personnel transport: Transportation of personnel to and around work sites.
- Hardware transport: Transportation of hardware components to work sites or secondary base components.
- Production: Import of raw materials for production modules.
- Leisure: Excursions to interesting surface features.

Figure 11
Lunar Outpost's unpressurized LTV



14.0 LANDING AND LAUNCH VEHICLES, SITE LOCATION

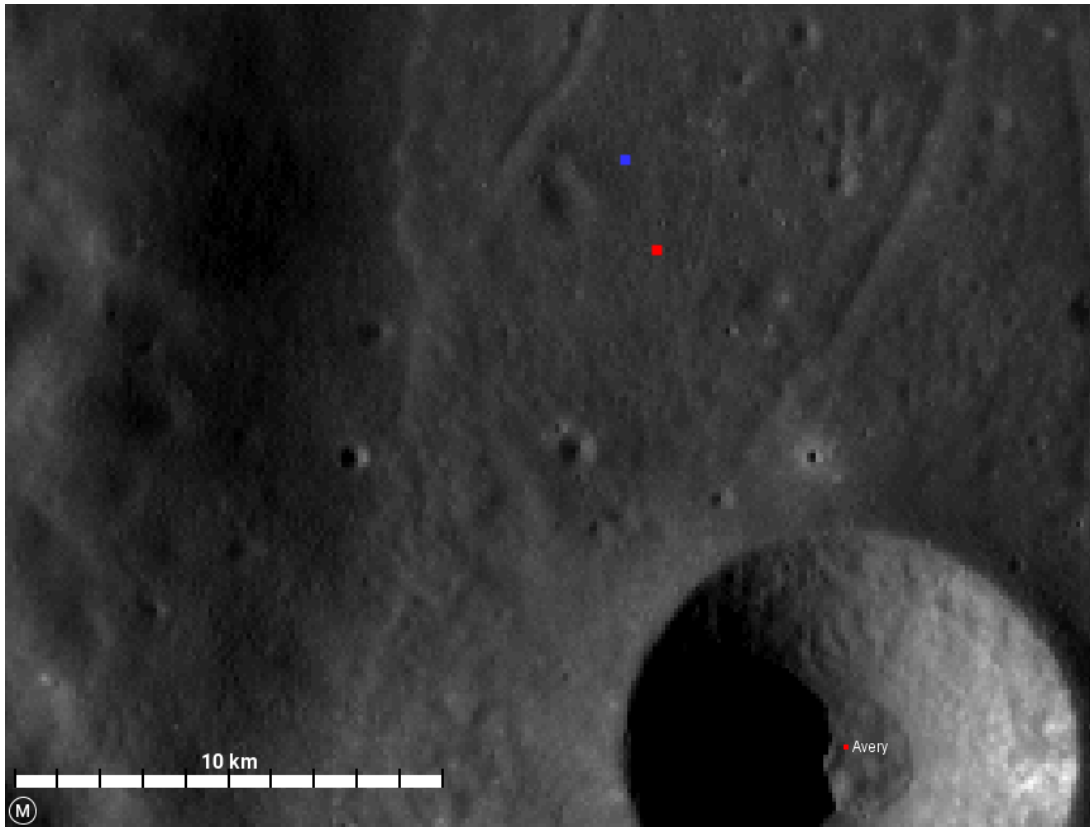
Moonbase AndrewMills-1 will make use of SpaceX's Starship (figure 12) for crew, heavy cargo launch, and landing capability. It has a heavy lift capacity, necessary for a large crew. It also uses a fuel/oxidizer combination of methane and liquid oxygen (LOX) that could be produced in-situ, using the Sabatier reaction in the methane production module for the methane and the O_2 extracted from the lunar soil for the LOX.

Figure 12
SpaceX Starship Human Landing System



Launch and landing will take place from a flat region 2 km to the northwest of the base at 0.864 deg S 81.196 deg E shown in figure 13. The pad area will be compacted by rovers prior to astronaut landing to help with dust mitigation. The distance from the base, along with berms dug by robotic excavators (repurposed from the shielding excavators) will reduce the reach of the dust cloud. The launch site should not be located any further for emergency evacuation's sake, as 2 km is considered an upper limit of EVA walking distance.

Figure 13
Launch/Landing location (blue) next to base location (red)



15.0 LUNAR BASE ISRU

In order for Moonbase AndrewMills-1 to complete its mission, supplies will be needed for 6 months of continuous operation. Part of this mission will be to prove the ability of a moon base to operate with a degree of independence from the Earth, so some of those supplies will be produced from on the surface. These are the basic needs of a manned moon base:

- Oxygen:
 - Supply to maintain constant perfusion throughout the base and pressurized rover.
 - Pressurized form as an EVA consumable.
 - Liquid form for use as a propellant oxidizer.
- Water:

- Daily astronaut needs.
 - For use in KRUSTY reactors.
- Food
 - Daily astronaut needs.
- Miscellaneous
 - Radiation shielding.
 - Landing pad.
 - Manufacturing feedstock.
 - Crop growth in lunar soil.

15.1 ISRU at Moonbase AndrewMills-1

Due to the composition of Mare regolith shown in figure 14, these are the possible specific outputs of ISRU efforts, and the reactions by which Moonbase AndrewMills-1 will obtain these outputs:

- Oxygen: Considering the significant concentrations of SiO_2 found in the Pyroxene, Olivine, and Anorthite of the Mare regolith, a molten electrolysis method (figure 15) will be used for the O_2 extraction module. This method will allow for a high volume of O_2 production.
- Water: With an equatorial location, lunar water would not be abundant at Mare Smythii, so ISRU of water will not take place
- Iron: The molten electrolysis method (figure 15) employed by the O_2 extraction module will also be capable of 2Fe production.
- Titanium: 2TiO_2 will be produced as a byproduct of the molten electrolysis reactor, which will be separated into titanium powder by using an aluminothermic reduction reaction to isolate the titanium ($\text{TiO}_2 + 4/3\text{Al} \rightarrow \text{Ti} + 2/3\text{Al}_2\text{O}_3$). (Sciencedirect.com)
- Aluminum: The leftover $2/3\text{Al}_2\text{O}_3$ from the aluminothermic reduction reaction, along with $2\text{Al}_2\text{O}_3$ found in Pyroxene will be separated into 2Al and O_2 using the Vapor Pyrolysis Process (figure 15).
- Lunar concrete: Lunar concrete ingredients would be similar to those of terrestrial concrete: an aggregate, water, and cement (Physical properties of concrete made with Apollo 16 lunar soil sample). Water would need to be brought from Earth. Calcium aluminate cement

could be produced from lunar Anorthite. Lunar regolith could be used as the aggregate. This method was demonstrated using a sample of Apollo 16 regolith.

Figure 14
Mare regolith average concentration

MARE REGOLITH

Ilmenite - 15%

$\text{FeO} \cdot \text{TiO}_2$	98.5%
---------------------------------	-------

Pyroxene - 50%

$\text{CaO} \cdot \text{SiO}_2$	36.7%
$\text{MgO} \cdot \text{SiO}_2$	29.2%
$\text{FeO} \cdot \text{SiO}_2$	17.6%
$\text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	9.6%
$\text{TiO}_2 \cdot \text{SiO}_2$	6.9%

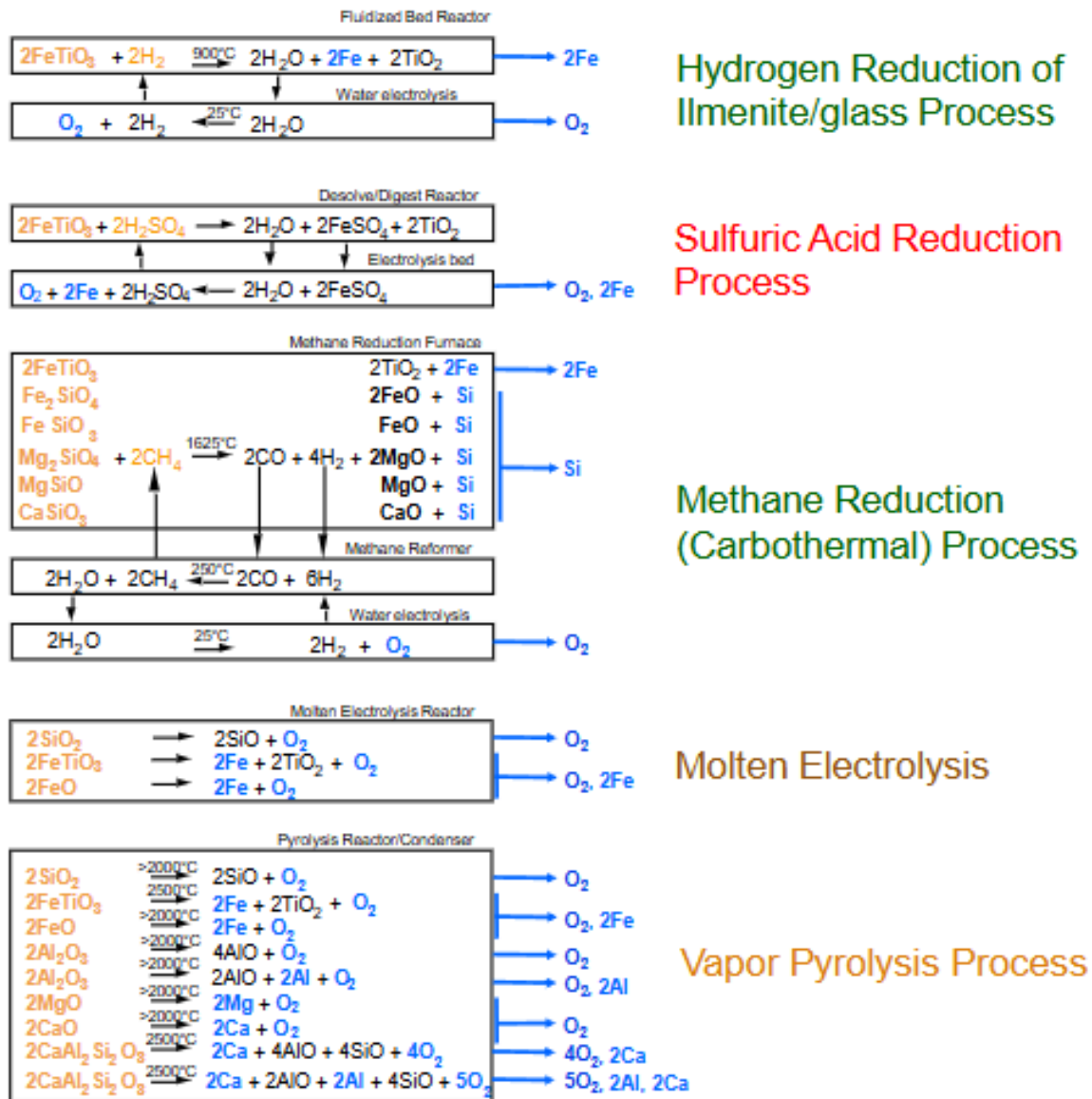
Olivine - 15%

$2\text{MgO} \cdot \text{SiO}_2$	56.6%
$2\text{FeO} \cdot \text{SiO}_2$	42.7%

Anorthite - 20%

$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	97.7%
---	-------

Figure 15
Various applicable reactions and their outputs



15.2 ISRU Expected For Mission Success

The goal of ISRU at Moonbase AndrewMills-1 is to prove it as a long-term concept, and learn how these ISRU systems change over an operational lifetime. Mission success will come from these lessons learned, but there will also be production goals associated with each ISRU product:

- O₂: 10000 kg would be a realistic production goal with a 2 kW molten electrolysis reactor operating continuously for 6 months. This wouldn't be enough to fully refuel the starship lander, but would help significantly while also covering astronaut needs many times over.
- Building supplies (Iron/Titanium/Aluminum/Lunar concrete): Should be produced at a level necessary to construct basic architecture concepts in order to prove its feasibility as a future cheap lunar construction material.

15.3 Types and Locations of ISRU Modules

There will be two ISRU modules used in Moonbase AndrewMills-1:

- O₂ extraction module: Houses a 2 kW molten electrolysis reactor for use in O₂ and Fe production. Located on the eastern side of the south wing of the base.
- Regolith processing plant: Houses an aluminothermic reduction reactor and vapor pyrolysis reactor for Ti and Al production. Anorthite from regolith will also have its calcium aluminate separated for use as cement in this module. Located at the western side of the south wing, just before the west wing.

16.0 MOONBASE ANDREWMILLS-1 FINAL LAYOUT

The layout of Moonbase AndrewMills-1 is shown in figure 3.

16.1 North Wing

The north wing will be the clean work/living area for the astronauts. This will be a shirtsleeve environment, and to pass into the south wing there will be a short cleanroom procedure. Here is the layout of the north wing in a clockwise order:

- Habitat A
- Medbay

- Habitat B
- Bio Lab
- Control Center
- Waste Management
- Galley
- Recreation Center

16.2 West Wing

The west wing will be a clean work area for astronauts, covering the EVA and high water-use modules. This will be a shirtsleeve environment, but the water recycler modules would be a high-noise environment, so hearing protection PPE should be worn in and near these areas. Here is the layout of the west wing in north to south order.

- Airlock A
- Greenhouse
- Water Recycler A
- Water Recycler B
- Airlock B

16.3 South Wing

The south wing will be the dirty work area for the astronauts, covering the entire processing flow of lunar regolith from import to product use. Astronauts will have to wear dust mitigation and hearing PPE in the south wing, and go through short cleanroom procedures to enter the north or west wings. Here is the layout of the south wing in a clockwise order:

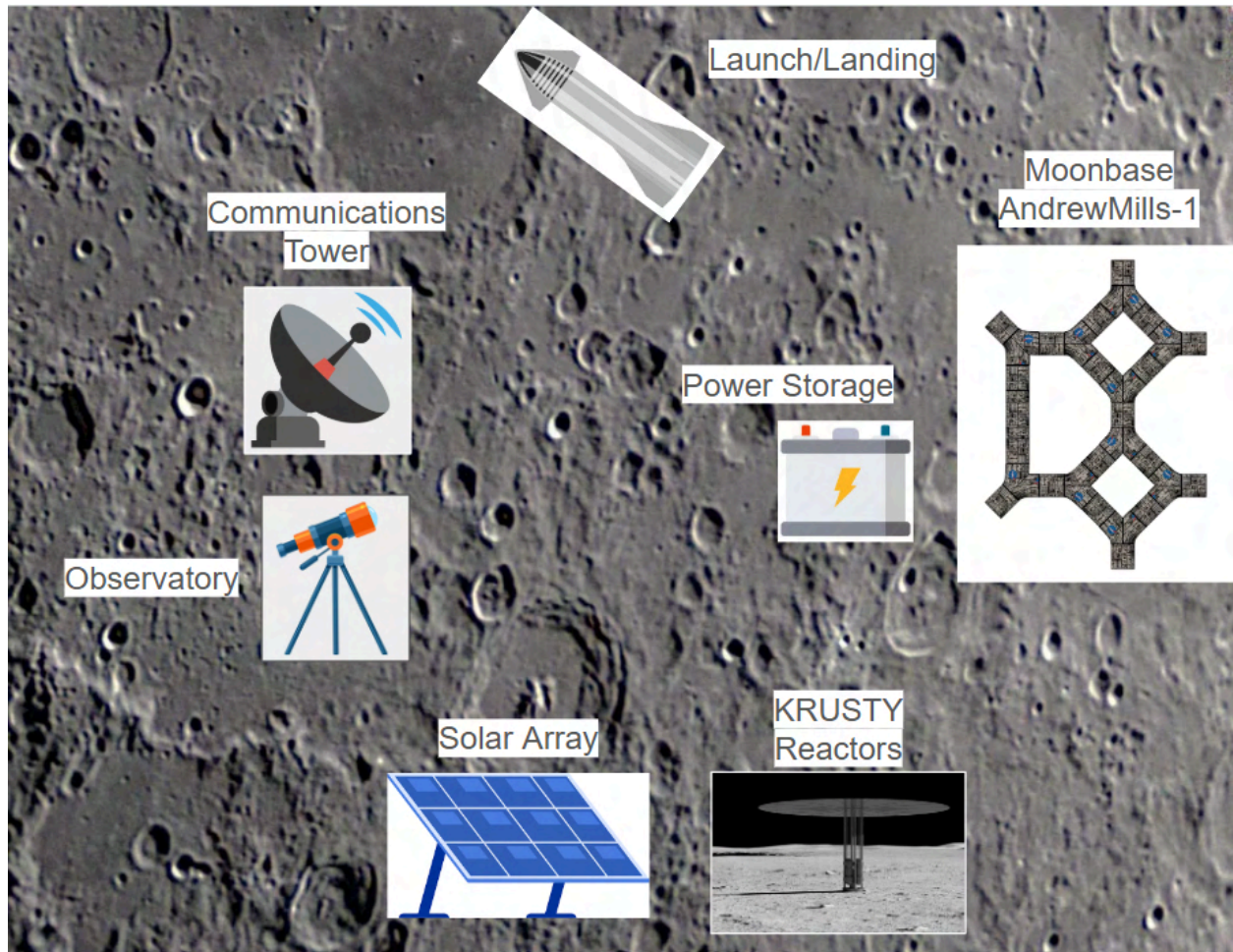
- Methane Production
- O₂ Extraction
- Fuel Depot
- EVA Garage
- Storage
- Regolith Processing Plant

- Selenology Lab

16.4 Nearby Modules

Some of the modules of the base will be located outside of the pressurized, radiation shielded section, and will therefore be inaccessible without EVA. The layout of these modules is shown in figure 17. The solar array and KRUSTY reactors are located to the southwest of the main base, far enough to avoid the dust plume from the launch/landing area (1 km). Power distribution cords carry power from these modules to the power storage module, and from there to the base, communications array, or observatory array. The energy storage module will be located adjacent to the southwest corner of the base for quick EVA servicing. The communications and observatory arrays will be located along the west side of the base, near enough for easy servicing. The launch/landing pad will be located 2 km northwest of the base, far enough to avoid major dustup.

Figure 17
Layout of the modules nearby to Moonbase AndrewMills-1



17.0 LUNAR BASE SUMMARY

Moonbase AndrewMills-1 will be located near the equator along the eastern limb of the Moon within Mare Smythii. This location is within a Mare region, and is close to both highlands and deep craters, promoting exploration of diverse environments. All habitable base elements will maintain a line of sight to the Earth at all times, making high-density communication simple. Temperature extremes, radiation, and meteorites will be mitigated by a regolith shield, keeping astronauts protected in a shirtsleeve environment. Science modules will allow astronauts to conduct meaningful experiments, while greenhouse and hydroponic modules will supply fresh food and greenery. The remaining modules of the base

support science, ISRU, and astronaut health. The base power will be generated by an array of solar panels and KRUSTY nuclear reactors. Crew waste will be recycled as efficiently as possible, removing the need for resupply missions over the six month mission period.

The six member crew will have access to a culturally diverse menu, psychological support, and a virtual reality “holodeck” to help to ease feelings of isolation from Earth. Lunar Terrain Vehicles, one pressurized and two unpressurized, will aid astronauts in surface mobility, enabling excursions over a far-reaching area of the Moon. Selected crew will come from diverse scientific and engineering backgrounds, and will be trained in the soft skills necessary for strong working relationships. The medical astronauts and medbay module will be capable of supporting class I and II medical issues, but class III medical events will necessitate a launch back to Earth.

SpaceX’s Starship will ferry the base components, LTVs, and astronauts to the surface. Robotic surface assets will place the base modules, connect them together, and bury them in 2m of regolith. Support of the base would be a very involved process; each astronaut will need tasking information and support staff to complete their base functions. The production modules, nuclear reactors, observatories, and gardens will also need monitoring and control.

As a part of the Moon base design process the class participated in our own lunar experiences to better understand the difficulties faced during extraterrestrial survival. Here is a list of those activities:

- Moon astronomy: Take pictures of the moon as if you are on the moon taking pictures of Earth. Identify moon geographic features.
- Greenhouse simulation: Grow “crops” in a controlled environment, and observe their life cycle.
- Footprint simulation: Choose a lunar regolith simulant and try to recreate the famous Apollo footprints while observing characteristics of the simulant.

- Cratering simulation: Toss garden rocks of various sizes at a tub of lunar regolith simulant and study the ejecta patterns.
- Landing dust plume simulation: Release pressurized air over lunar regolith simulant to test how it would react to a thrust plume. Also tested different strategies for mitigation of dust agitation, such as plastic landing pads or compacted regolith.
- EVA simulation: Don ECLSS gear (heavy clothes, helmet, gloves, backpack) and perform various EVA tasks, such as picking up rocks with tools or stepping through tight doorways.
- Lunar rover activity: Drive an RC car over a bed of regolith simulant, seeing where challenges arise while driving over craters and difficult terrain.
- Regolith deposition activity: Bury an aluminum can in a regolith simulant using different proposed deposition methods to see which methods are easiest.
- Lunar landing and launch piloting: Use a moon landing simulator and choose between telemetry sources you would want in front of you during that landing.

With a significant financial contribution to NASA by congress or some benefactor, a base following a similar design to Moonbase AndrewMills-1 could begin construction within the next ten years, with astronaut operations beginning within two years of the beginning of construction.

APPENDIX A: SIMULATION OF LUNAR ACTIVITIES

A1.0 MOON 101

A1.1 Intro to the Moon

- Design considerations
 - Surface temperature varies ~250 K
 - Synodic period 29.5 days (closer to 15 days light and dark)
- Two main types of terrain:
 - Maria (Flat to gently rolling darker plains)
 - Highlands (Rugged, cratered, lighter)
- Cool info
 - Earth's albedo may be enough light for astronauts to work in.
 - Apollo 14 astronauts struggled to judge distance, creating a significant navigation problem for traversing highland environments.

A1.2 The Lunar Environment

- Design considerations
 - GCRs, solar wind, and SPEs are potential space weather events attenuated by lunar atmosphere. SPEs in particular could quickly cause loss of life and need consideration.
 - Lunar gravity field is lumpy, significantly challenging low orbits.
 - Low surface thermal inertia means that thermal effects can be avoided a few meters sub-surface
- Cool info
 - Earth and Moon are considered to be a binary planet!
 - Astronauts consider low gravity on the Moon an easier work environment than weightlessness
 - There are deep moonquakes timed with tidal forces with the Earth.

A1.3 Brief history of Apollo

- Design considerations
 - Apollo 11 showed that judging distance is significantly more difficult than expected.
 - Apollo 12 was able to land with 13m of accuracy to the target ellipse center.
 - Selenologically trained astronauts saved missions that would have otherwise had bad science return (Apollo 16) due to unexpected surface selenology.
 - Apollo 16 also showed that the highlands are a selenological mess.
- Cool info
 - Life threatening lack of mobility in pressure suit during Gemini 9 EVA
 - Dave Scott and Jim Irwin faked a seatbelt issue to go grab a chunk of basalt

A1.4 Future Science on the Moon

- Design considerations
 - Habitat lab analysis enables high grade sample collection and site revisits, as opposed to shorter sortie missions.
 - Mobility is the key to larger scale sampling missions, pressurized rovers could enable that.
 - Robots could and should be used for reconnaissance, and to pave the way for humans.
 - Automated deployment would save significant astronaut EVA time, but machines should be accessible and fixable! Jury rigging is not a long term solution.
- Cool info
 - MMOD is not considered too significant of a hazard (at least by some constellation scientists)

A2.0 ASTRONOMY FROM THE MOON

As a part of our lunar science simulation, we took pictures of the moon to learn its selenography and find the challenges associated with

observing features on another heavenly body. Figure A1 shows my observation of the moon using a camera and tripod setup. Figure A2 shows my observation of the moon using a telescope and phone camera setup.

Figure A1

Photo taken on 01/16/2025 using a camera-tripod setup on a clear night.



Figure A2

Photo taken on 03/14/2025 using a phone camera through a telescope during a lunar eclipse.



A3.0 GREENHOUSE SIMULATION

As a part of our lunar surface science simulation, I chose four seeds to grow in my home “greenhouse”. I chose rocket salad, peas, moonflowers, and zinnias. To have a good sample size, I chose to grow three seeds of each type in an empty egg carton, as shown in figures A3 and A4. I watered each seed twice a day.

After one week, the rocket salad and zinnias began to sprout (figure A5), and grew slowly and in great numbers for the following three weeks. Two of the moonflowers and only one of the peas were able to sprout, and they grew much taller than the rocket salad and zinnias over the next three weeks. After four weeks had passed, the rocket salad and zinnias grew too

tall and started to fall under their own weight, and quickly died. The moonflowers and pea stalk remained in good health up until week five when they started to droop, dying by the end of the week.

This experiment gave a good lesson in the need to transfer seeds as they use up nutrients in soil, as these seeds remained in their seed-starter soil for the entire duration of the experiment, and depleted that soil before blooming.

Figure A3
Empty egg carton seed starter setup



Figure A4
Filled egg carton with seeds planted at week 0



Figure A5
Picture taken at week 1 showing the first sprouts of the rocket salad



Figure A6

Picture taken at week 3 03/21/2025 showing all sprouted plants thriving



Figure A7

Picture taken at week 5 04/09/2025 when the last plant died



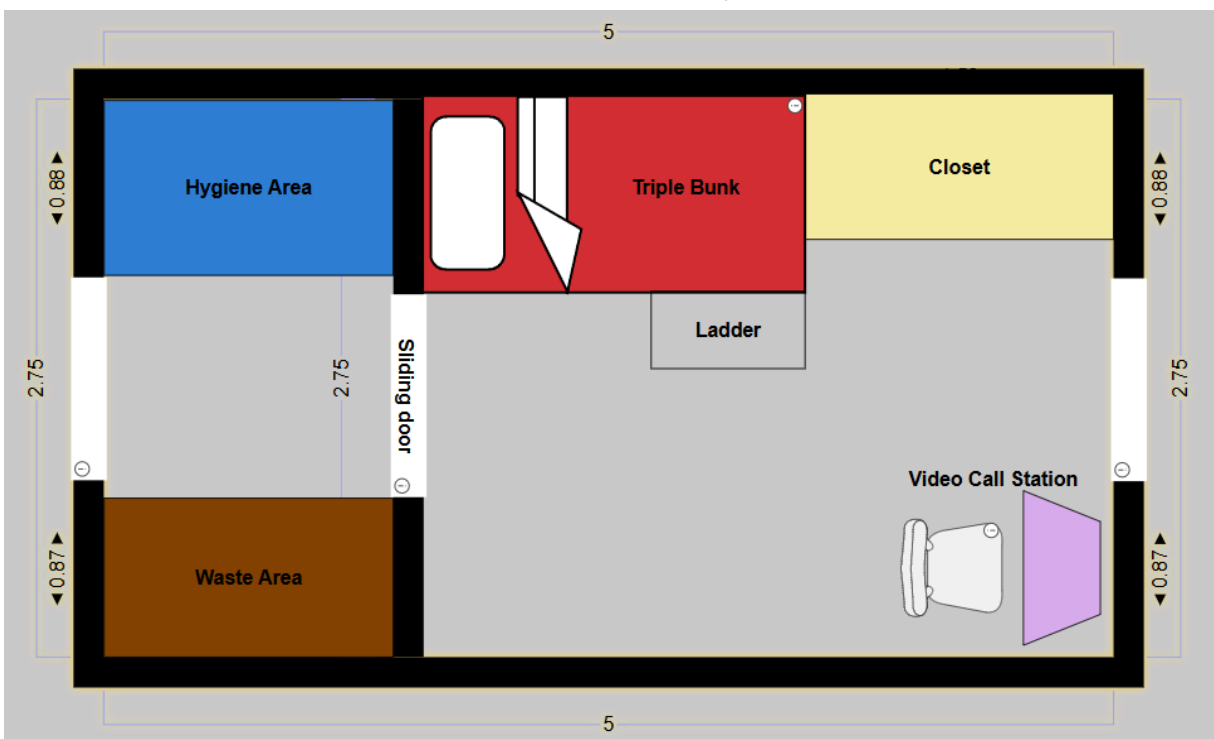
A4.0 MODULE LAYOUTS

A4.1 Habitat Modules (Figure A8)

A triple twin-size bunk was chosen to maximize floor area, allowing crew to use the extra available space for discretionary use. A ladder is used to access upper bunks. A separate area is sectioned off for bathroom and hygiene use. A closet will hold clothes for each of the three astronauts in each hab module. A video call station allows for constant contact with family. All walkways conform to the 1m requirement. Modules fitting this floorplan will be:

- Habitat A
- Habitat B

Figure A8
Habitat module layout

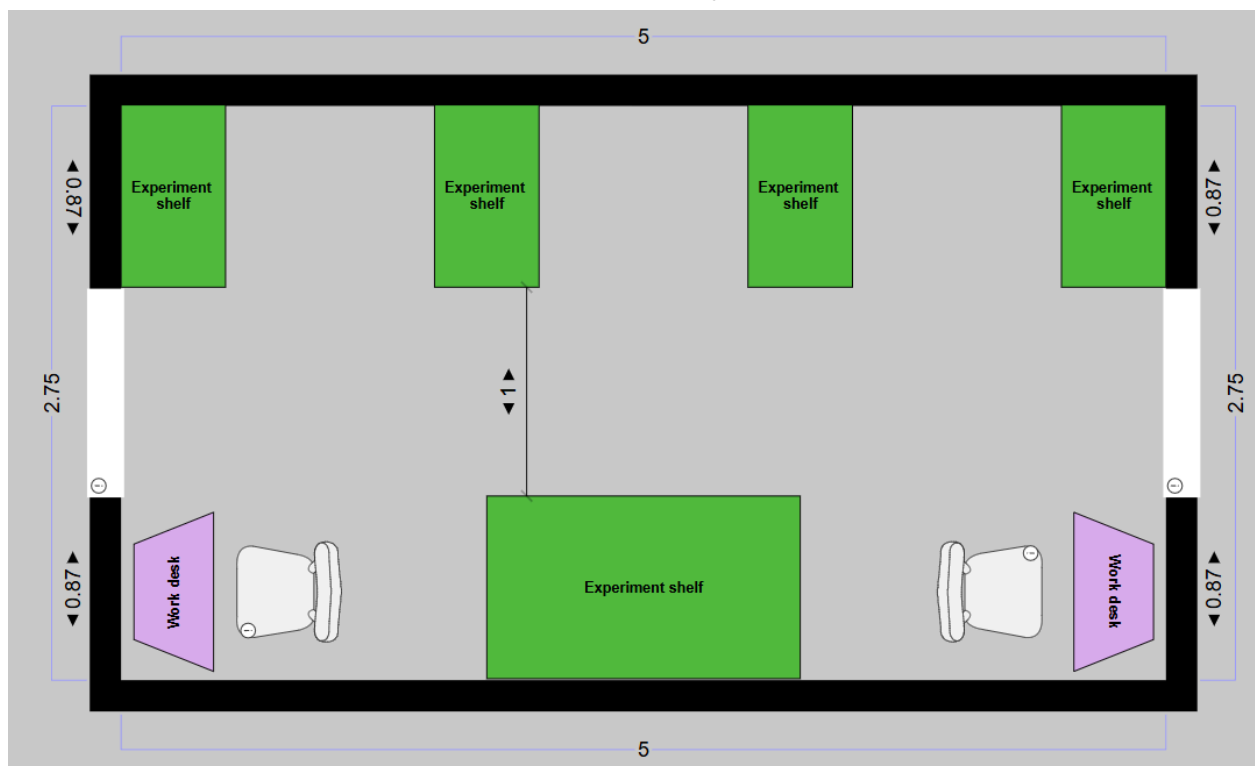


A4.2 Lab Modules (Figure A9)

This layout was chosen to maximize working space while still allowing for free movement. Each experiment shelf will have wheels, allowing access to different sides of the shelf. Each lab will support two working astronauts. Modules fitting this floorplan will be:

- Biology lab
- Selenology lab

Figure A9
Lab module layout

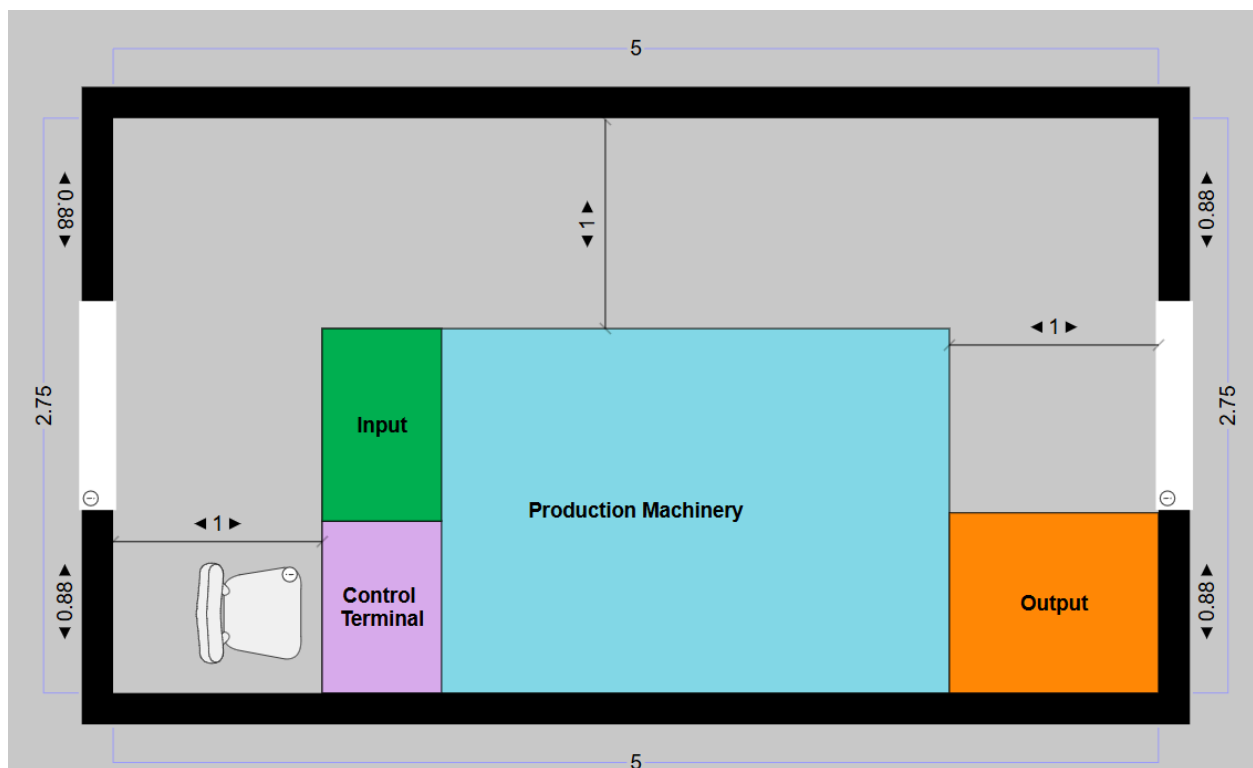


A4.3 Production Modules (Figure A10)

Production modules have been laid out to allow ample room for passage, but also maximize space for the production machinery. Terminals allow for production system control and observation, and input and output sections of the machinery are located near the exit hatches for easy loading and unloading. Modules fitting this floorplan will be:

- Water recycler A
- Water recycler B
- Waste management
- Methane production
- Regolith processing plant
- O₂ extraction

Figure A10
Production modules

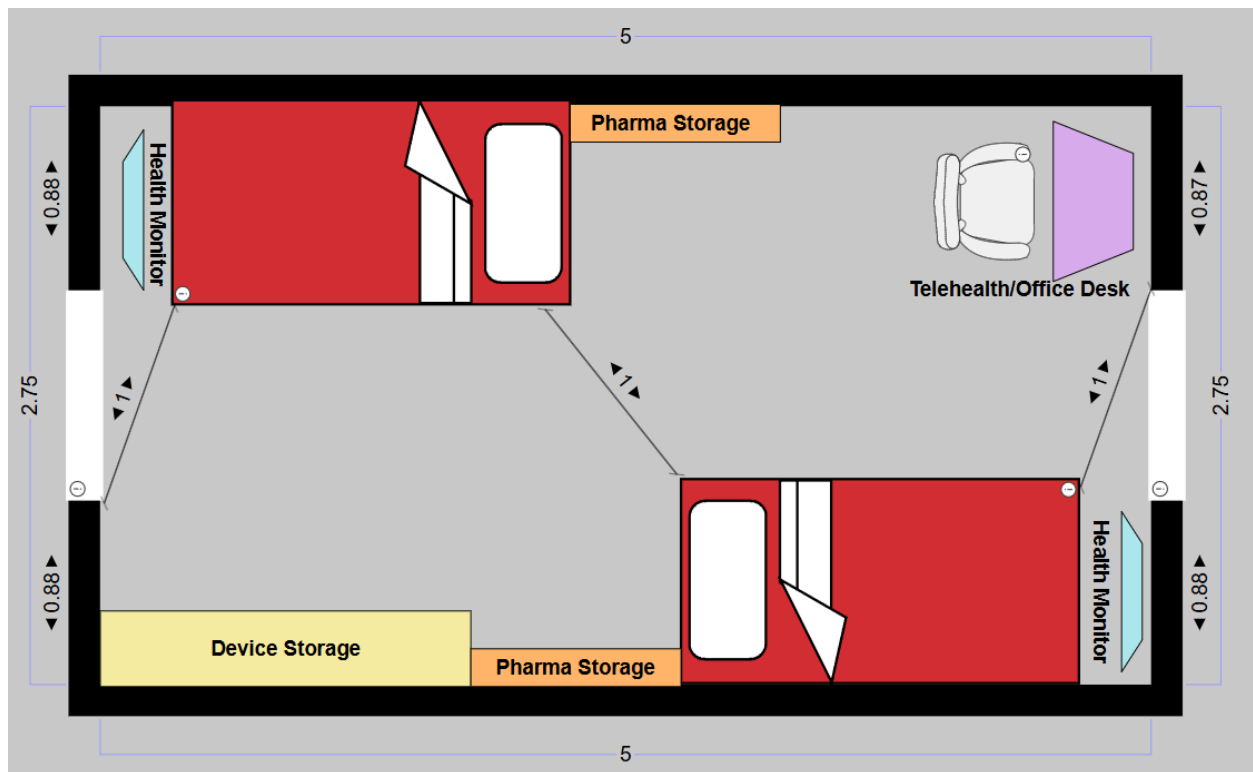


A4.4 Medbay Module (Figure A11)

The medbay will have space enough for two beds, storage space for medical devices, and shelves for pharmaceuticals. There will also be a mobile desk setup for telehealth appointments, medical note-taking, or mobile storage of clean medical instruments. Monitor screens will display required vitals for easy access to information. All walkways conform to the 1m requirement. Modules fitting this floorplan will be:

- Medbay

Figure A11
Medbay Module

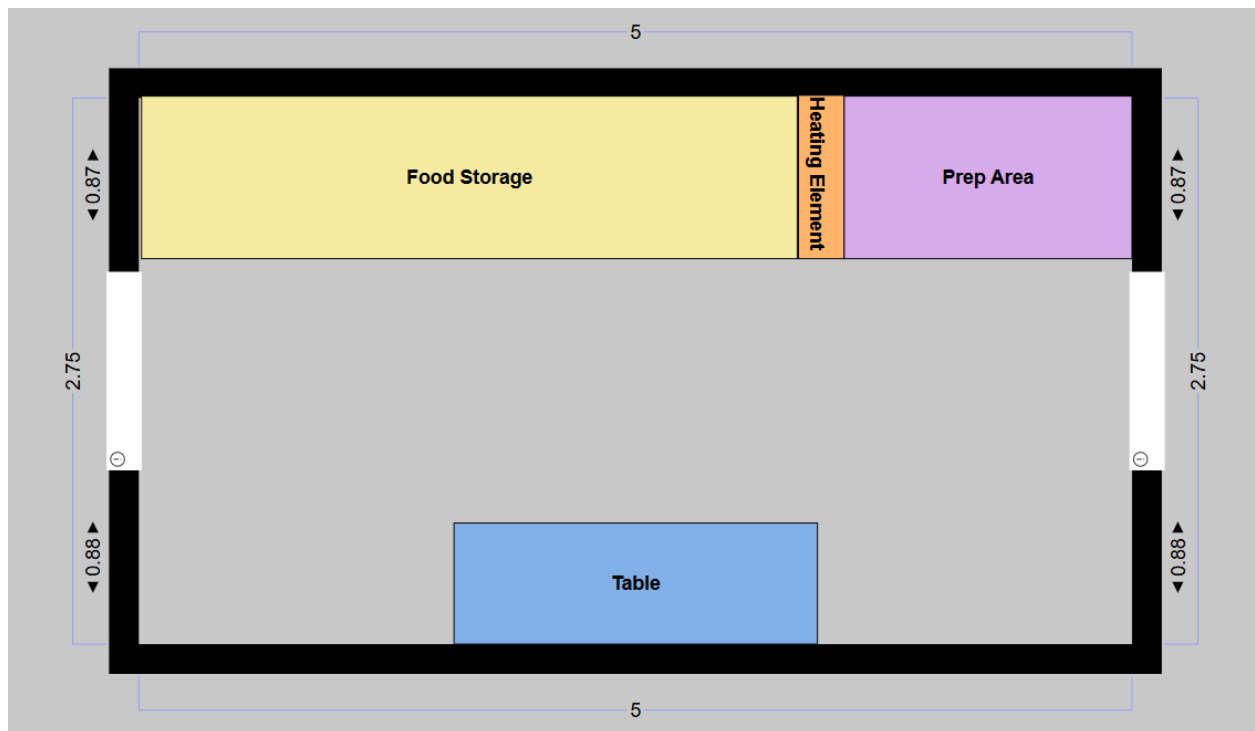


A4.5 Galley Module (Figure A12)

The galley module will have a large area setup for food storage, a small heating element, and a separate area for food preparation. The storage area will include space for perishable fresh food from the greenhouse/hydroponics, as well as for longer lasting provisions. There will also be a pull-out table with room to comfortably seat 6 astronauts so that group meals will be possible. Modules fitting this floorplan will be:

- Galley

Figure A12
Galley Module

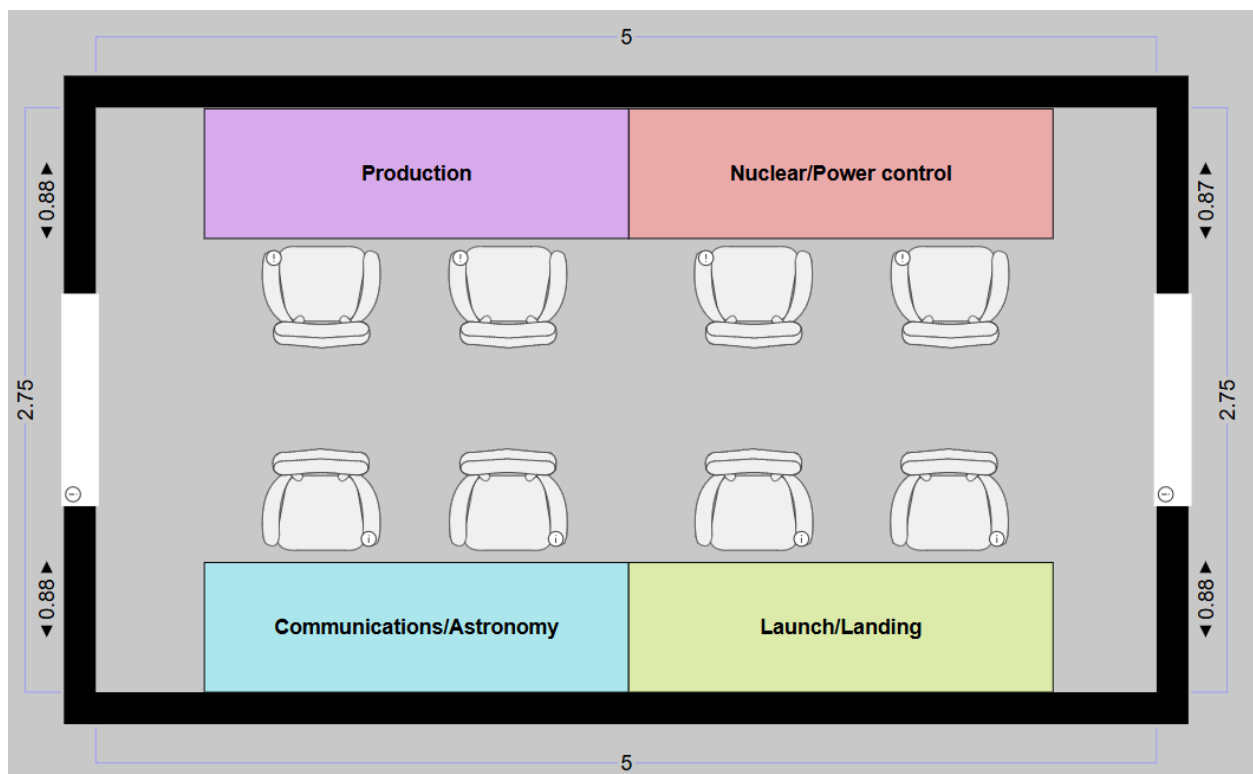


A4.6 Command & Control Module (Figure A13)

The command and control module will have four stations for local control of each major outpost operation. Most control should be relegated to Earth, but for the sake of an emergency or as a Mars analog, the capability should remain with the moonbase to locally control the production modules, nuclear power station, power distribution, communication systems, astronomy suite, launch, and landing. Each station will be capable of control by one or two astronauts. Modules fitting this floorplan will be:

- Control center

Figure A13
Command & Control Module

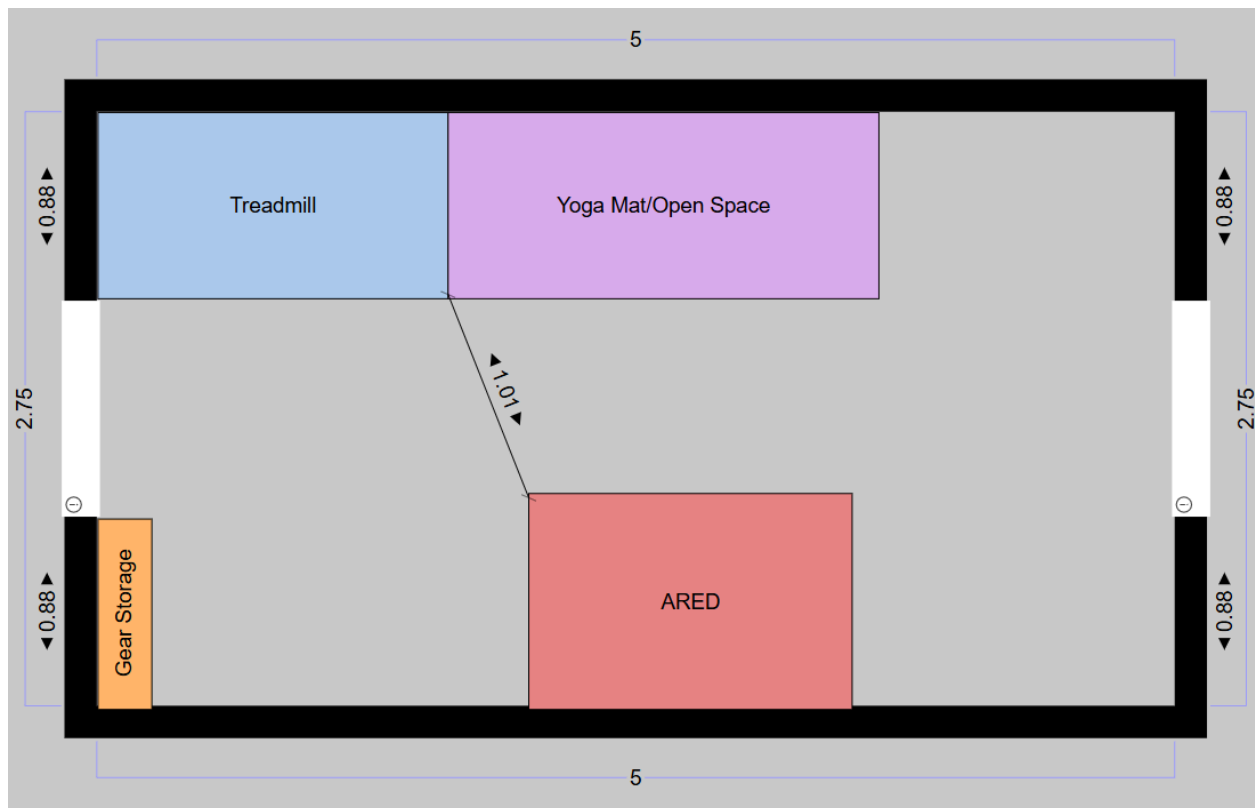


A4.7 Recreation Module (Figure A14)

The recreation module was designed to meet the constant exercise needs of astronauts on long-duration missions. Much like the ISS, there will be a treadmill and ARED-style resistive training machine, but since gravity plays some part a yoga mat and stretching area can also be added. This module will also serve as the home of a VR holodeck-type system, proposed at the Humans-to-Mars 2024 conference, where astronauts can maintain their skills in a virtual environment, interact with loved ones in a 3D space, or practice for upcoming complicated EVA procedures. Modules fitting this floorplan will be:

- Recreation center

Figure A14
Recreation Module

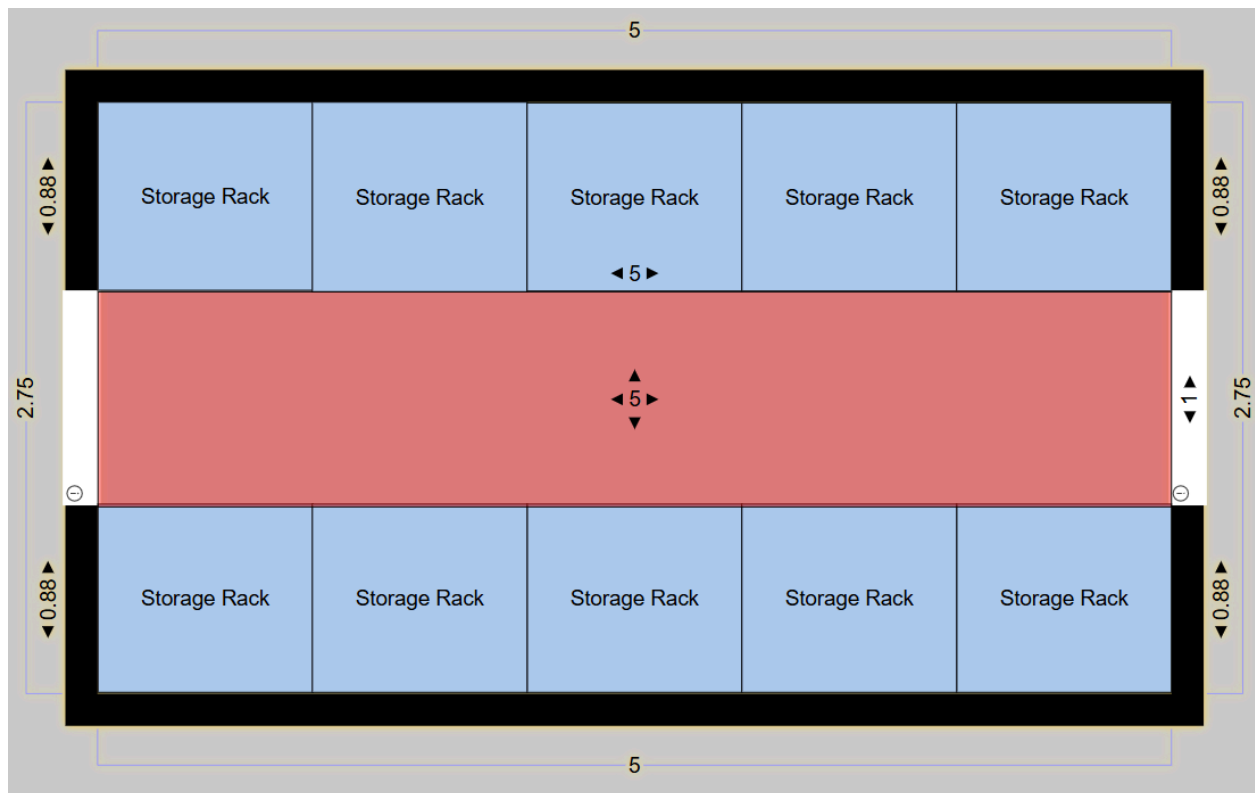


A4.8 Storage Module (Figure A15)

The storage modules will serve as a place to organize and separate items that need storage. 10 separate storage racks will enable adequate separation of regolith types, tools, spare parts, spacesuits, clothes, or other items that aren't needed day to day. Modules fitting this floorplan will be:

- Storage
- Fuel depot
- EVA garage

Figure A15
Storage Module



The requirements for free space in a shirtsleeve environment of 206 m² are met by a large margin; uses should be found for the abundant free space in the Y joint modules. Possible uses include hydroponic shelves or extra storage.

A4.9 Airlock Modules (Figure A16)

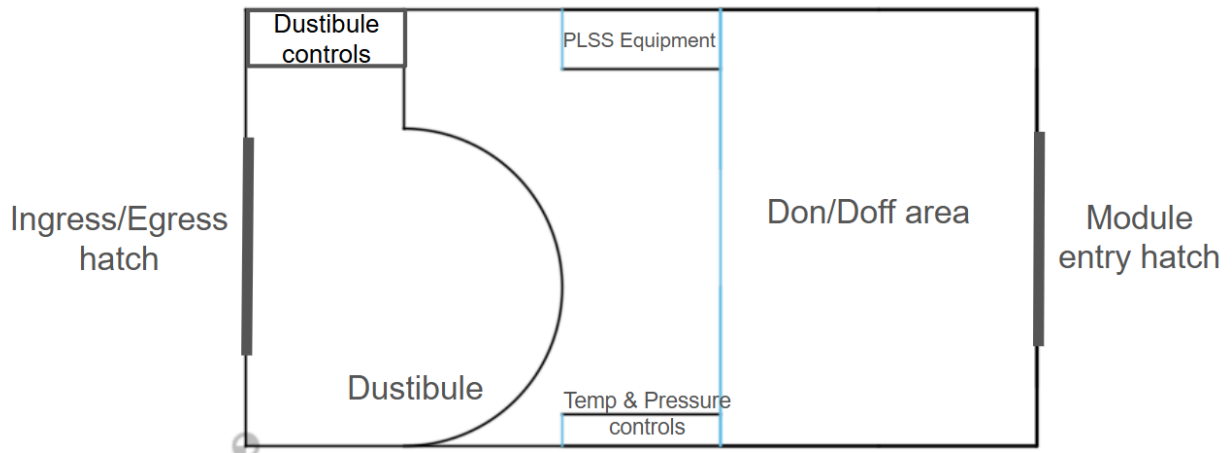
Moonbase AndrewMills-1 will have two airlocks. They will each be located at the corner of the Y-joint passages facing out from the west wing. Both will have the same design. Hatches will be 1.016m by 1.524m to allow EVA suits to pass through with minimal sealing length along the door. They will include:

- 8 cubic meters at least for two astronauts to don/doff EVA suits per NASA's 2018 guidance.
- An area for dust cleaning of EVA suits, or “dustibule”, which would look like a car wash without the water. There's a proposal for lunar regolith “swiffer dusters”, which will be employed by astronauts to perform finishing touches after using the dustibule.
- Control panels to adjust the temperature and pressure of the airlock between the base temperature and pressure towards the surface temperature and vacuum.
- Storage for:
 - EVA suits
 - PLSS canisters
 - PLSS batteries

Modules fitting this floorplan will be:

- Airlock A
- Airlock B

Figure A16
Airlock module layout



These airlocks will hold four EVA suits each so that two sets of two astronauts can leave from one airlock without needing to transport an EVA suit through the base (which would be possible). This would leave two extra suits just in case there's damage sustained in the suits over time.

Free area: Light red sections in figures A17-A26 give free floor area numbers for the functional modules and joint modules, I sum those numbers for the total free floor space. I then multiply that number by 3.5 m (the ceiling clearance) to obtain the free volume. To fill out the totals in the following table, I then multiply each module-specific total by the number of modules fitting that floorplan.

Table A1
Free space available within modules

Module	Free floor area (m2)	Free volume (m3)
Habitat (2)	6.88	24.08
Lab (2)	8.82	30.87
Production (6)	7.23	25.305
Medbay	7.62	26.67
Rec Center	10.62	37.17
C&C	8.88	31.08
Galley	8.61	30.135
Storage (3)	5	17.5
I Joint	3.255	11.3925
Y joint	28.343	99.2005
L Joint	11.049	38.6715
Total	384.117	1344.4095

Figure A17
Lab module free floor space

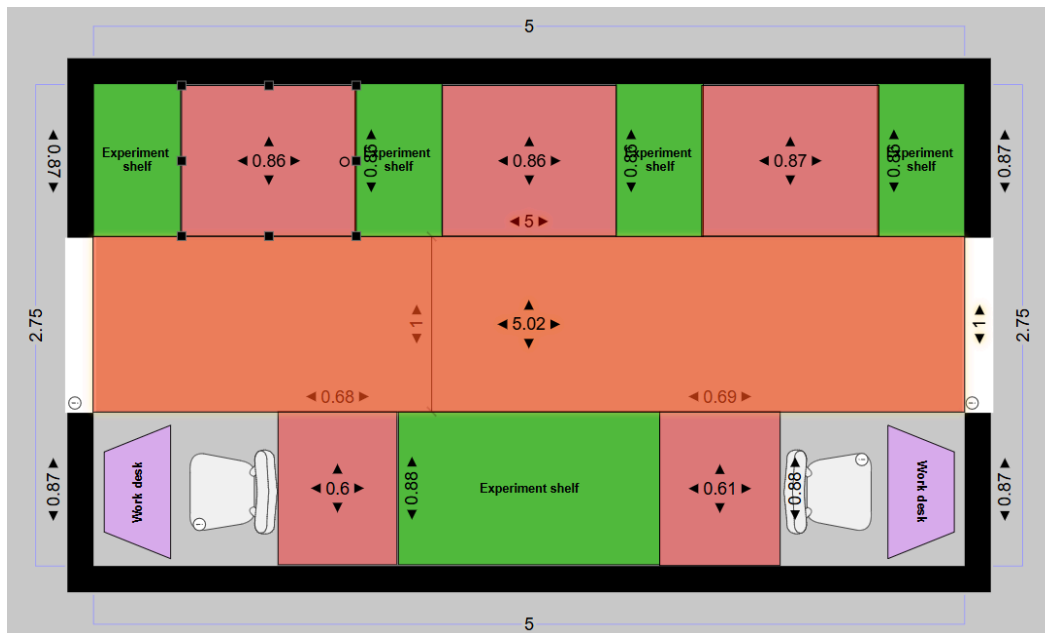


Figure A18
Hab module free floor space

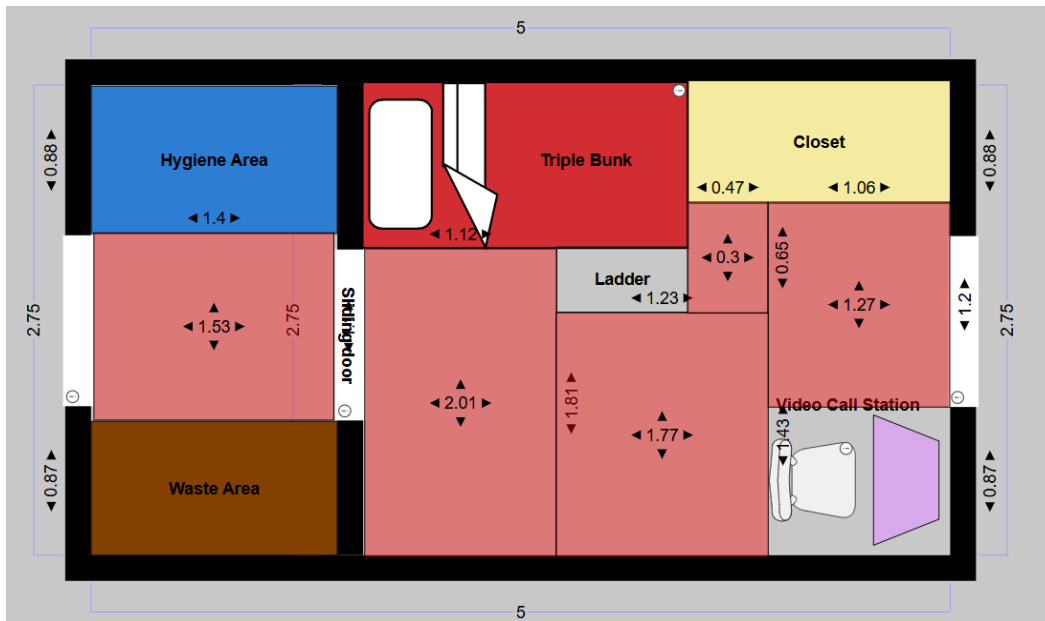


Figure A19
Production module free floor space

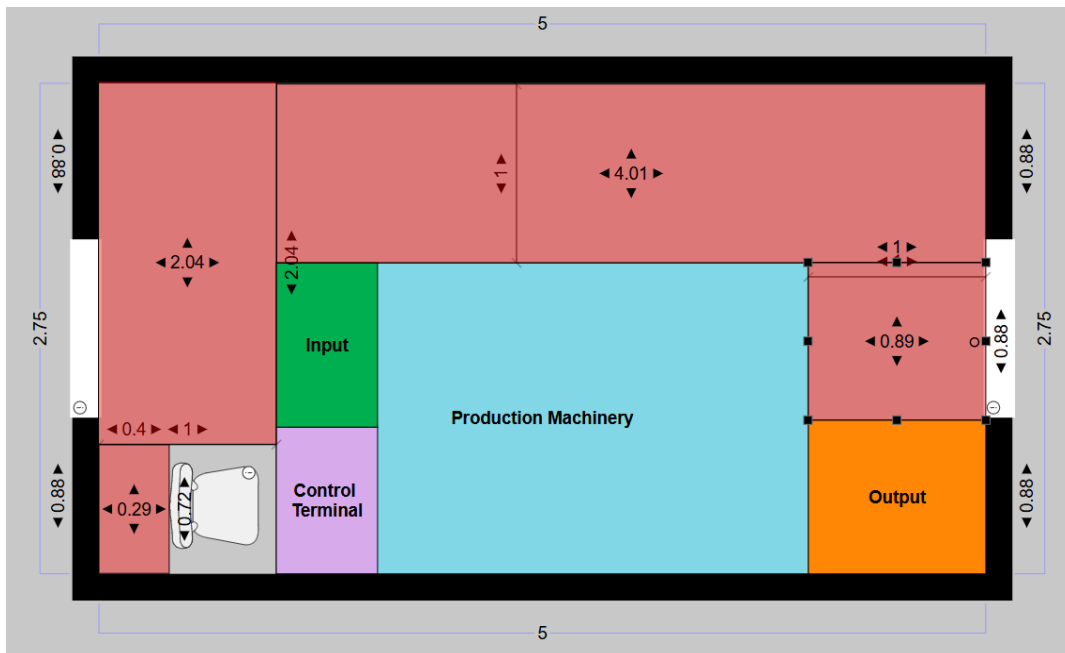


Figure A20
Storage module free floor space

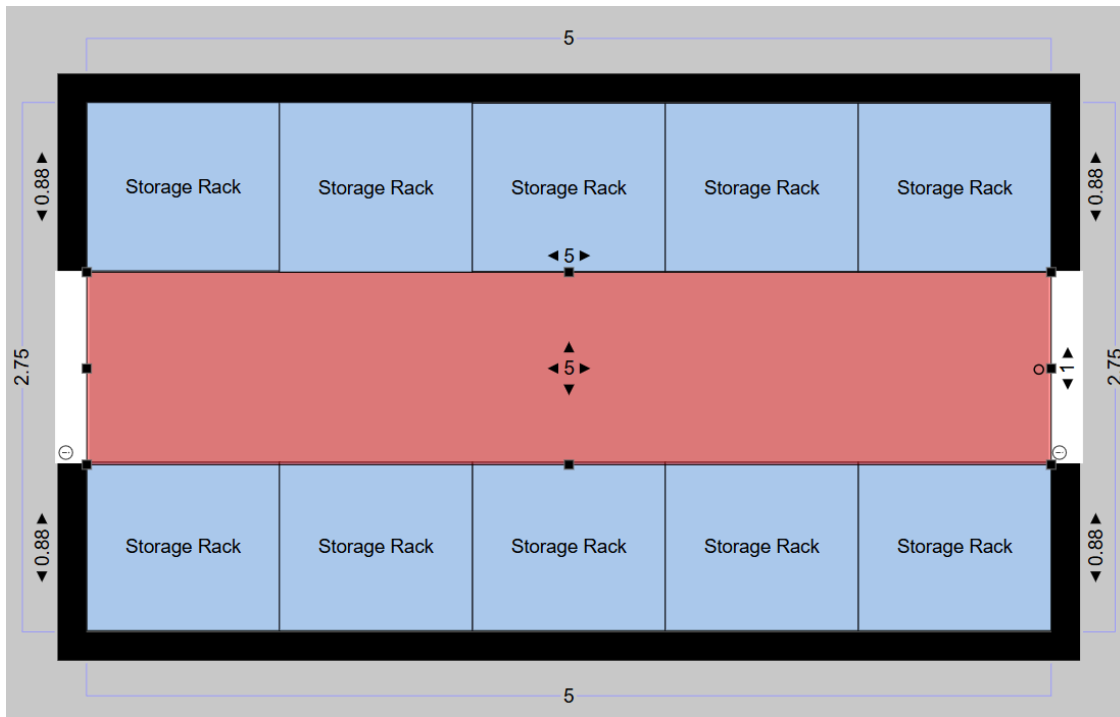


Figure A21
Command and Control module free floor space

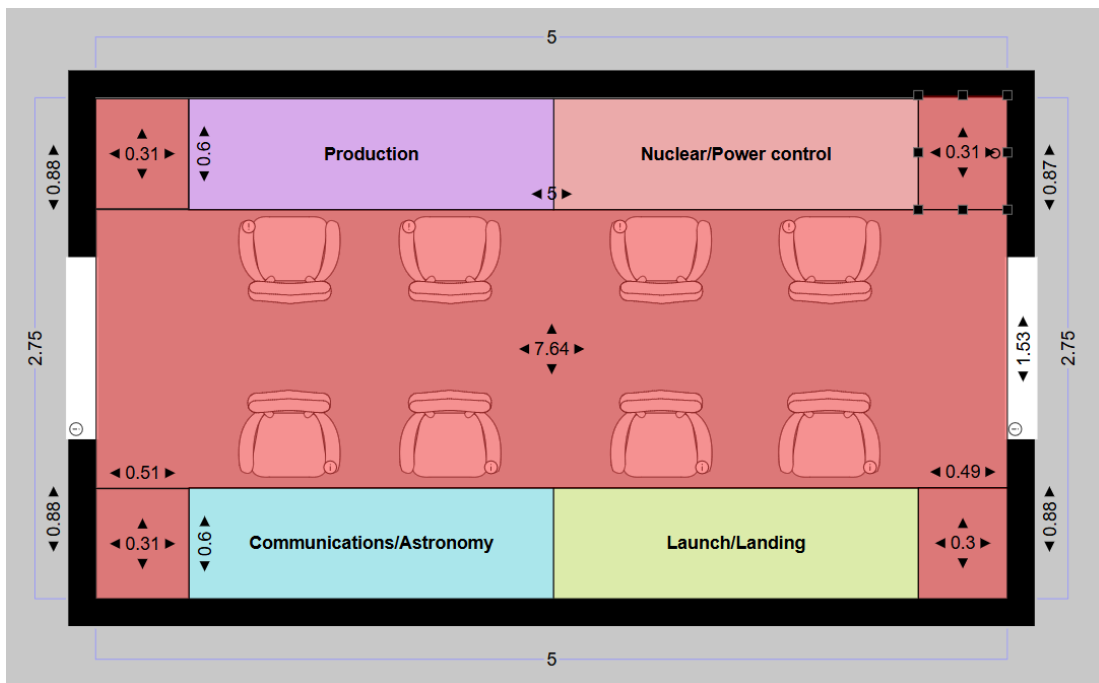


Figure A22
Medbay module free floor space

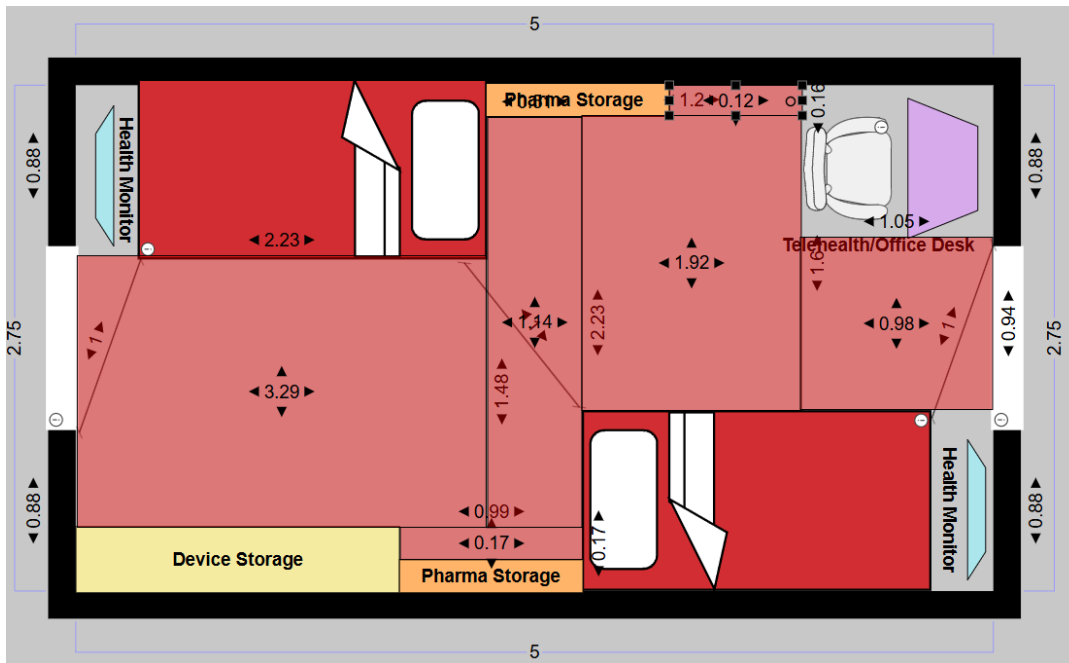


Figure A23
Recreation module free floor space

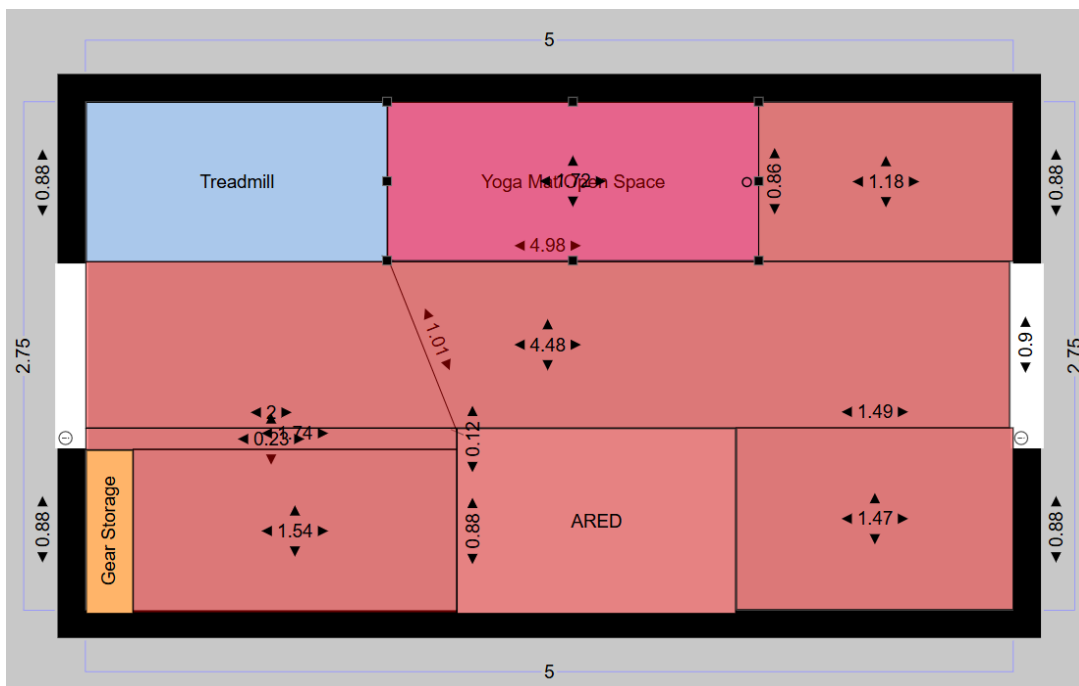


Figure A24
Y joint module free floor space

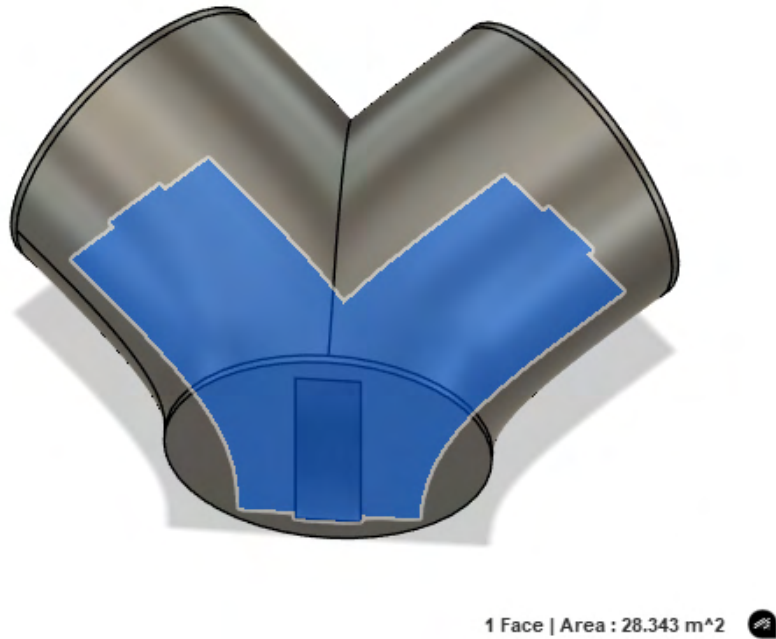


Figure A25
L Joint module free floor space

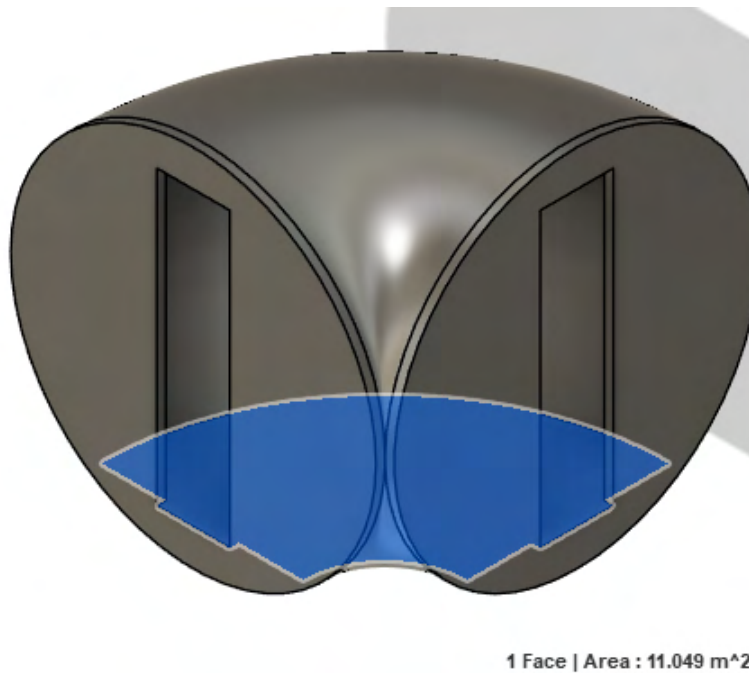
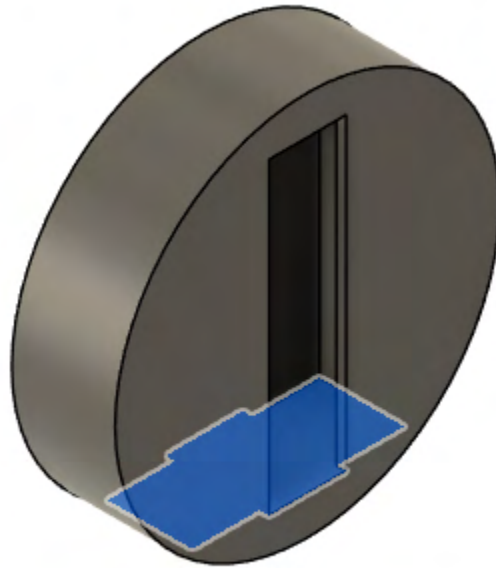


Figure A26
I Joint module free floor space



1 Face | Area : 3.255 m²

A5.0 LUNAR FOOTPRINT SIMULATION

Footprints were applied to flour as an analog for moon soil, and observations were taken of how they held shape and the difficulty of creating the footprints.

- When a toy astronaut was used, the footprints only held shape if we were very gentle (either compacted or loose). (Figure A27)
- Loose flour allowed for deeper boot prints, but didn't look like Apollo boot prints. (Figure A28)
- Compacted flour held boot shape well and looked like Apollo boot prints, but were shallow (even with a jump). (Figure A29 and A30)

Figure A27
Souvenir clay bootprints



Figure A28
Loose flour boot prints



Figure A29
Making a compacted flour boot print



Figure A30
Compacted flour boot prints



A6.0 LUNAR CRATERING SIMULATION

The flour was covered with a thin layer of cocoa powder so that any ejecta from the impacts would contrast with the rest of the surface. The flour was not compacted before cratering. Four trials were conducted, the first with small rocks (figure A31), then medium (figure A32), then large (figure A33), then medium at an angle (figure A34). Small rocks showed little to no ejecta. Medium rocks created a small local crater with thin linear ejecta patterns emanating from that crater. Large rocks created deep craters with random ejecta patterns reaching further than the linear medium rock patterns. The angled medium rocks only sent ejecta in the direction of velocity, and usually skipped off the surface.

Figure A31
Small rock crater patterns



Figure A32
Medium rock crater patterns



Figure A33
Large rock crater patterns



Figure A34
Angled medium rock crater patterns



A7.0 LUNAR LANDING THRUST PLUME SIMULATION

To simulate dust plumes from landing in the lunar environment, we lowered a handheld air pump slowly to the surface of loose flour. We attempted this experiment in three ways: loose flour, loose flour covered by a solid landing pad (tupperware lid), and compacted flour. In the case of loose flour, a huge dust plume was created (figure A35), and coated a large amount of the room with flour. In the case of a landing pad, minimal dust movement happened at all (figure A36), only a little as the air pump was very high above the surface. In the case of compacted flour, there was a small dust cloud formed when high above the surface (figure A37) and there was little dust while near the surface (figure A38), but there was a small blast of dust as the compacted flour fractured during touchdown (figure A39).

Figure A35
Loose flour close to the surface



Figure A36
Landing pad over loose flour



Figure A37
Compact flour (high above surface)



Figure A38
Compact flour (close to surface)



Figure A39
Compact flour (After touchdown)



A8.0 THERMAL ENVIRONMENT ON THE MOON

Using the latitude of the base site and a 180 degree range of sun angles, the surface temperature was calculated for the 14 day lunar day (figure A40).

Figure A40
Calculation for Tm at Mare Smythii

	A	B	C	D	E
1	Sun angle	Tm		lat	-0.932
2	0	0			
3	1	=373.9*(COS(RADIANS(\$E\$1))^0.25)*(SIN(RADIANS(A3))^0.167)			

A9.0 REGOLITH COVERAGE CALCULATIONS

Figure A41

Case 1: Retainer structure with 2m regolith shield

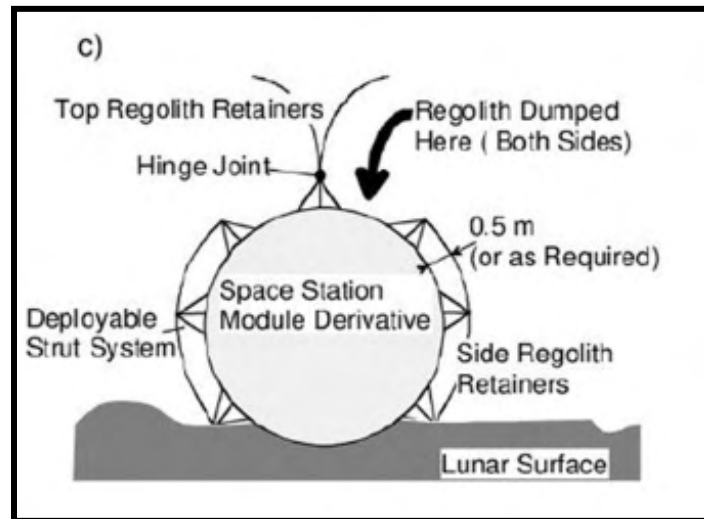


Figure A42

Case 2: Unsupported minimum mound diagram

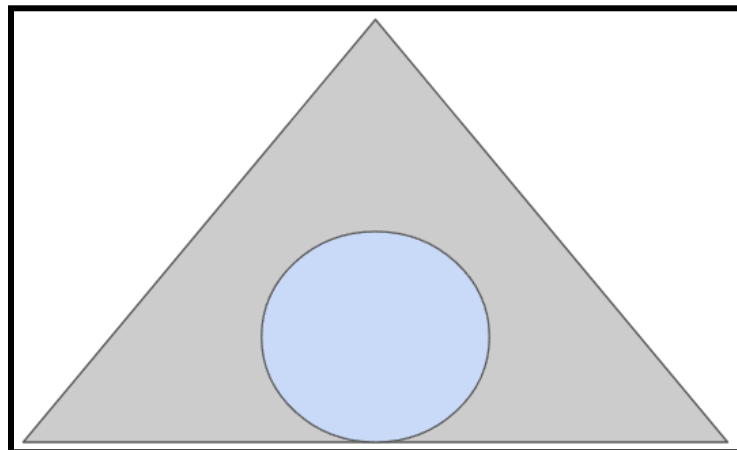


Figure A43

Case 3: Flat bin structure with 2m on each side and 2m on top

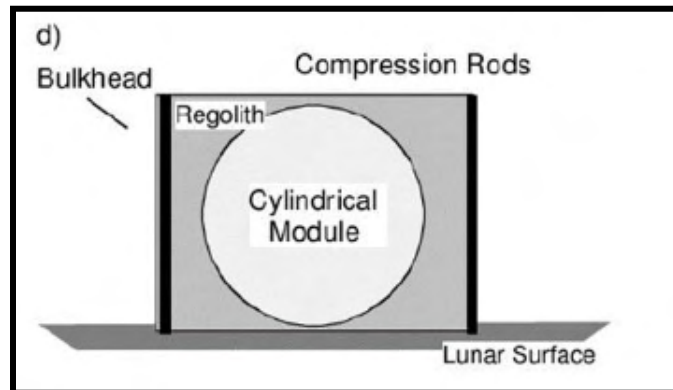


Figure A44

Case 4: Bin structure with 2m on each side until half way then 2m semicircle over the top

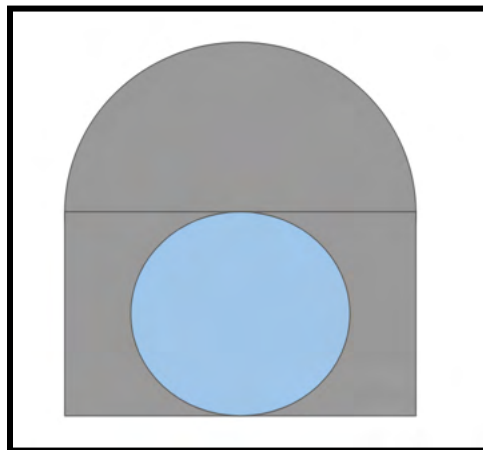


Table A2

Totals in each case and minimum regolith volume

Empty module volume	98.17477042
Case 1	219.9114858
Case 2	214.245
Case 3	216.8252296
Case 4	173.3683577
Modules	19
Minimum regolith volume	3293.998796

Figure A45
Unsupported minimum mound calculation

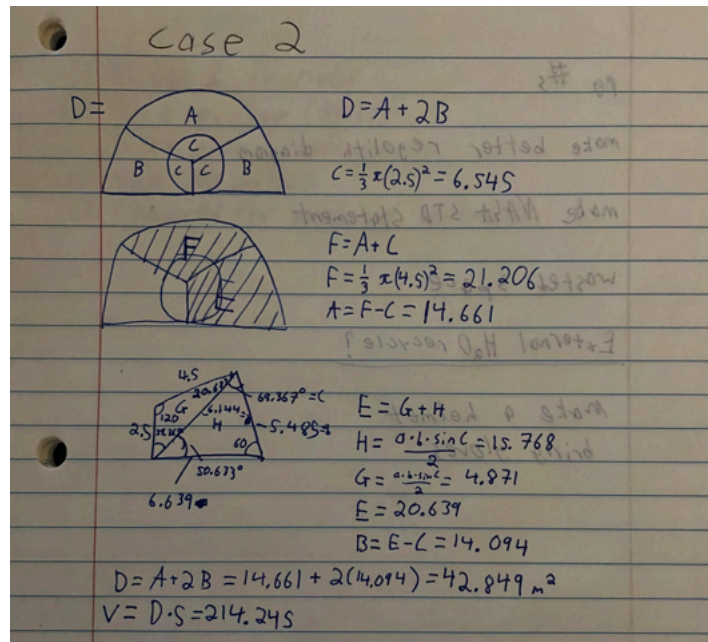


Table A3
Formulas for each case and total volume with minimum case

Empty module volume	98.17477042	
Case 1	$(\pi) \cdot 5 \cdot (9/2)^2 - B15$	
Case 2	$(42.849 \cdot 5)$	
Case 3	$(5 \cdot 9 \cdot 7) - B15$	
Case 4	$((\pi) \cdot 5 \cdot (9/2)^2) / 2 + (9 \cdot 5 \cdot 2.5) - B15$	
Modules	19	
Minimum regolith volume	$B20 \cdot B19$	

A10.0 LUNAR EVA SIMULATION

The procedure was:

1. Gather in lobby
2. 2 persons per EVA - one is prime second is helper, then reverse
3. Don the PLSS and visor and gloves
4. Gather tools and sample bags
5. Exit front EVA door – watch for step

6. Do mobility exercises per list
7. Collect rock samples – subsite traverse and selenological sampling tasks – see below
8. Photograph things
9. Walk to end of building
10. Go back in
11. Get vacuumed
12. Measure for Rover seat
13. Remove PLSS, etc

PLSS Gear (figures A49 and A50):

- Winter coat
- Snow boots
- Two layers of ski pants
- Pair of cotton gloves (figure A47)
- Ski gloves worn over cotton gloves (figure A48)
- Visor
 - Visor had a flashlight and camera taped to it to allow recording and sight in low light (figure A46).
- Safety goggles

Figure A46
Visor with flashlight and velcro for phone camera attachment



Figure A47
Cotton under-gloves



Figure A48
Ski gloves worn over the cotton under-gloves



Figure A49
Me in the full “Moon Suit”



Figure A50
Moon suit side-profile



CHECKLIST OF MOVEMENTS DURING MOONWALK

1. Single kneel and recover (figure A51)

Figure A51
Picking up a rock off of one knee



2. Subject picks up rock from ground using glove (figure A52)

Figure A52

Picking up Moon rocks wearing gloves (taken with helmet camera)



3. Subject picks up rock from ground using tool

The tool I used to pick up rocks was a small plunger-operated claw, capable of picking up small to medium sized rocks. (figures A53-A56)

Figure A53

Plunger-operated claw, claw portion demonstration

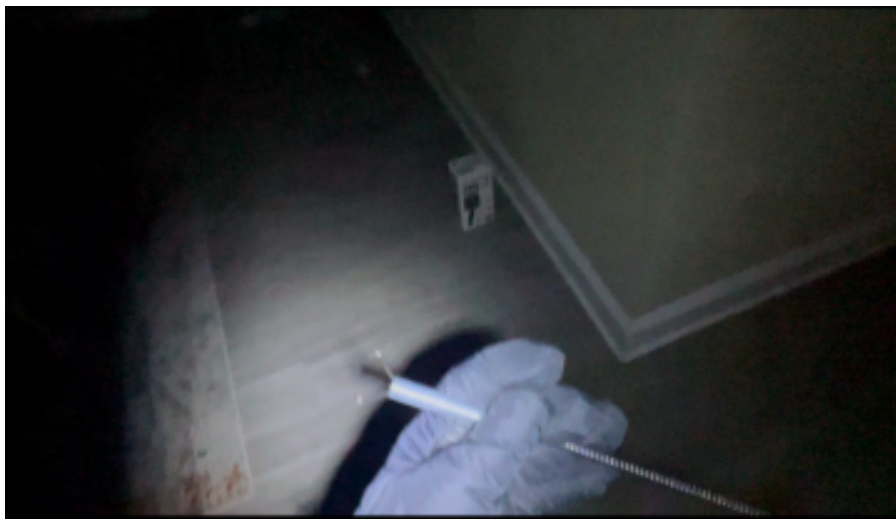


Figure A54
Plunger-operated claw, plunger portion demonstration



Figure A55
Claw picking up small rock from bent position

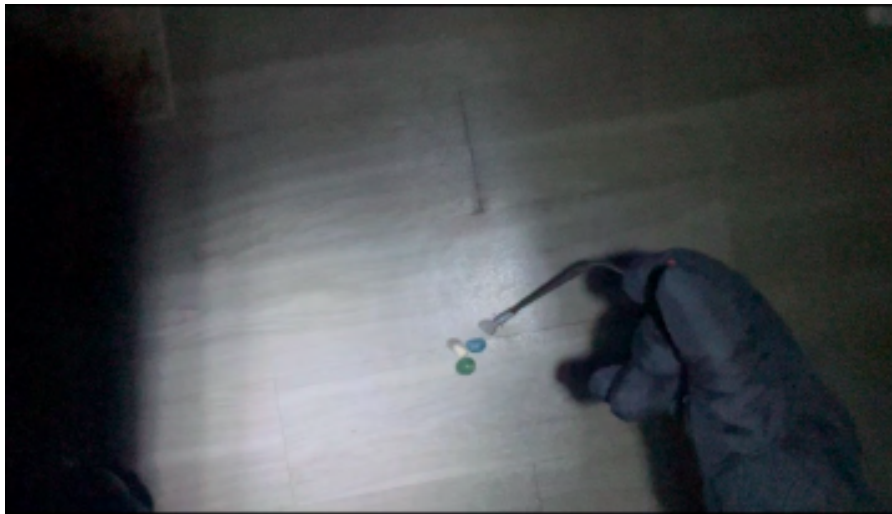


Figure A56
Bringing small rock up to other hand for collection



4. Subject walks on simulated moon surface

Walking was fine in my moon suit since each element is meant for Earth use, but wearing thick pants would have made walking for much longer uncomfortable. Visibility was low since I had taped a camera to the face plate, that could have been avoided by taping the camera to the side of the helmet.

5. Subject is measured for lunar rover seat with backpack on

- Measurement was:
 - Depth: 25 inches
 - Width: 21 inches

Observations from simulated lunar EVA

1. Ease of movements above

- Each of the movements was fairly easy, but was certainly more difficult than if I was wearing no moon gear at all. Wearing thick pants made kneeling down difficult. The glove combination made complex movement clumsy, writing my name came out

basically illegible. Basic hand movements in the gloves such as grabbing or pushing were unaffected.

2. Ease of using tools to pick up things

- Using the claw-plunger tool to pick up the rocks was clumsy and slow, and was made difficult by the low visibility of my helmet. It did make bending down unnecessary, which marginally helped.

3. Backpack space and movement

- To simulate a large backpack, I stuffed my old school bag with textbooks and clothes. This would make working close to other persons clumsy, as I needed to double check my pack wouldn't hit anything close to me before making any type of turn.

4. Walking on simulated lunar surface

- Walking was fine in my moon suit since each element is meant for Earth use, but wearing thick pants would have made walking for much longer uncomfortable.

5. Going through "airlock"

- Easy with my backpack, but would have been made more difficult with thicker pants.

6. Other observations

- Headlamp worked well, but with only one light source it was easy for a glove to get in the way of the rock I was trying to pick up, making me lose the rock in darkness. I would suggest stereo lighting to avoid this problem.

A11.0 LUNAR ROVER ACTIVITY

The goal of this activity was to show the challenges of operating a vehicle on surfaces similar to the lunar environment. These were the targets of the lunar rover activity:

- Drive rover (RC car) over simulated lunar surface (Flour)
 - Drive rover around the edge of simulated crater
 - Drive rover into and out of crater
 - Speed drive rover into crater and out
 - Drive rover until out of communications range

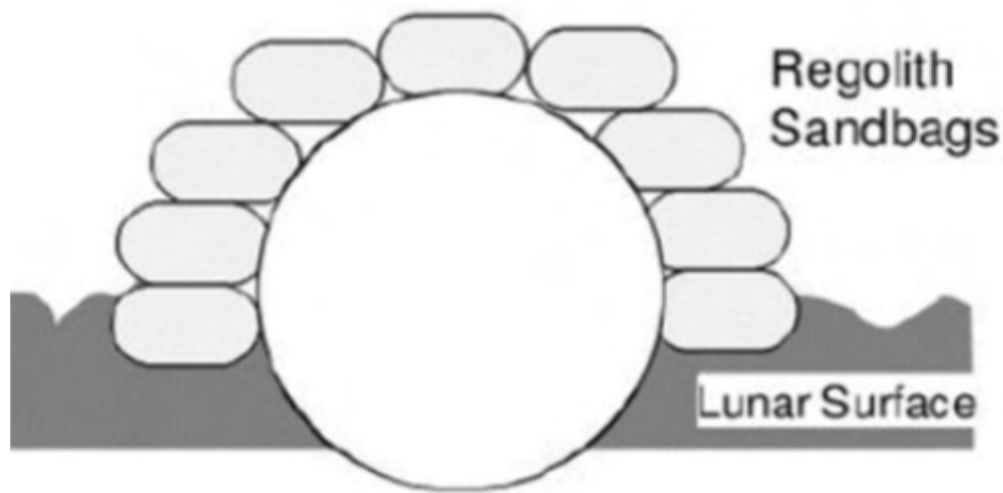
I wasn't present at the lunar rover activity, but one of the challenges observed was that there was difficulty getting over hills without speed. This could cause issues when astronauts try to overcome the lip of a crater, as excessive speed could lead to accidents.

A12.0 REGOLITH DEPOSITION

The goal of this simulation was to try out the various mechanisms of regolith deposition in order to find potential difficulties in this operation on the moon. The four cases to be tested for deposition are as follows:

- Lay regolith as in a) over lunar habitat - unsupported mound (figure A42)
- Lay regolith as in case b) - bulkhead (figure A43)
- Lay regolith as in case c) – half bulkhead (figure A44)
- Lay regolith bags over lunar habitat d) – sandbag (figure A57)

Figure A57
Case d: Regolith sandbags



I wasn't present at the regolith laying activity, so there are no pictures or observations.

A13.0 LUNAR BASE POWER NEEDS

Calculation of needed power – I estimated the power needed to run Moonbase AndrewMills-01 using the technical paper “Small Lunar Base Camp and in Situ Resource Utilization Oxygen Production Facility Power System Comparison”(Colozza 2020).

1. Calculation of solar array area (Table A4 & A5)
2. Calculation of nuclear power reactors (Table A4 & A5)
3. Calculation of battery storage size (Table A4 & A5)

Table A4
Values of power usage and generation

	A	B	C	D
1	OUTPUTS		INPUTS	
2	GaAs total area required	123.7362132	GaAs efficiency	0.32
3	DOE cell total area required	87.99019608	DOE cell efficiency	0.45
4	KRUSTY generated power	60000	Solar flux	1360
5	KRUSTY generators needed	6	Estimated total power	53850
6			Li-S gravimetric energy density	2510
7				
8	E generated per day (MWh)	38.2536		
9	Li-S battery mass (kg)	1524.047809		

Table A5
Formulas used to find power usage and generation values

	A	B	C	D
1	OUTPUTS		INPUTS	
2	GaAs total area required	$D5/(D4*D2)$	GaAs efficiency	0.32
3	DOE cell total area required	$D5/(D4*D3)$	DOE cell efficiency	0.45
4	KRUSTY generated power	$ROUNDUP(D5,-$	Solar flux	1360
5	KRUSTY generators needed	$B4/10000$	Estimated total power	53850
6			Li-S gravimetric energy density	2510
7				
8	E generated per day (MWh)	$((D5*14*24)+(B4*14*24))/1000000$		
9	Li-S battery mass (kg)	$B8*100000/D6$		

A14.0 COMMUNICATIONS LINK CALCULATIONS

The calculations for the communications LOS range are shown in tables A6-A9. Each tower has been given a height of 10m. The western ridge tower has a 1550m high vantage point over regions to the east of it, but only 10m to its west. The elevation profile of the base location & surrounding area is shown in figure A58 using data from the LOLA (Lunar Orbiter Laser Altimeter) instrument from the Lunar Orbiter spacecraft.

Table A6

Range (a) of the western ridge tower to the east

Western ridge tower (East LOS)	
r	1738000
h	1560
a	73610.50947

Table A7

Range (a) of the western ridge tower to the west

Western ridge tower (West LOS)	
r	1738000
h	10
a	5895.747055

Table A8

Range (a) of the base tower

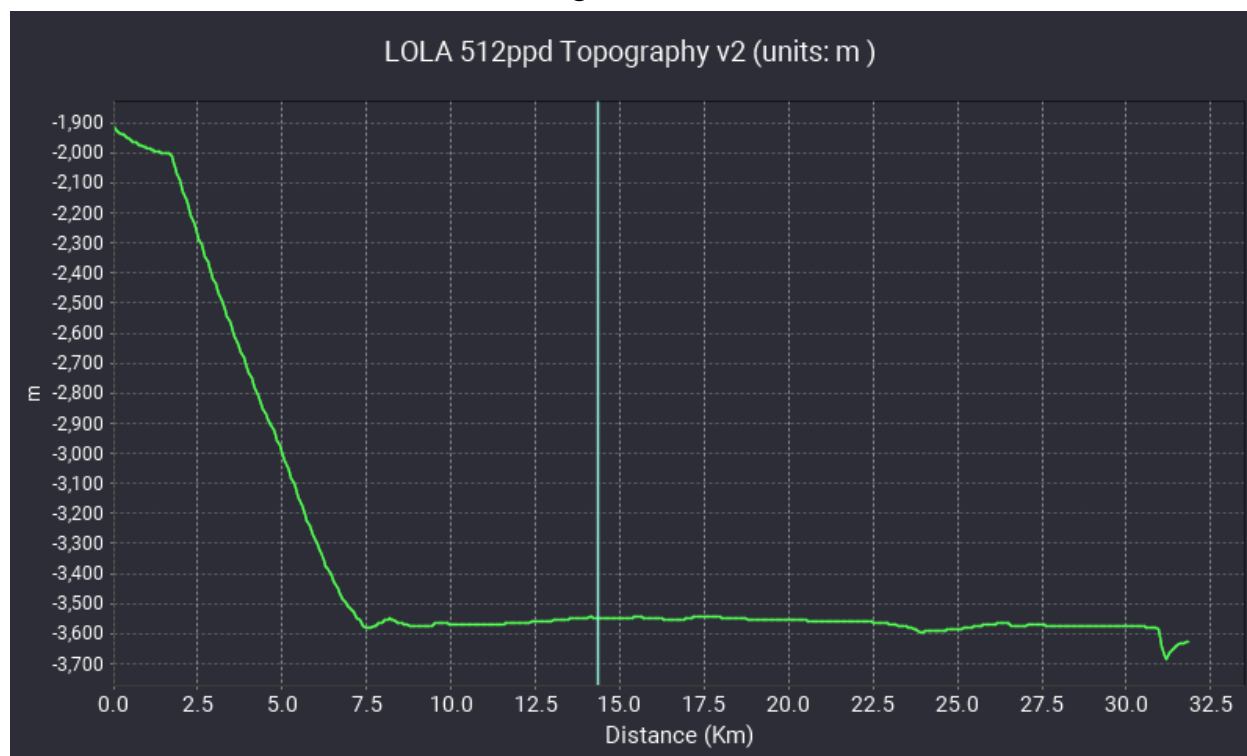
Base tower	
r	1738000
h	10
a	5895.747055

Table A9

Range (a) of the eastern mare tower

Eastern Mare tower	
r	1738000
h	10
a	5895.747055

Figure A58



A15.0 CREW SELECTION

To select a crew for Moonbase AndrewMills-1, the “right stuff” would be very different from what was selected for during the Apollo era. Soft skills will be just as important as any mechanical or scientific knowledge, and astronauts must be adaptable. That being said, on top of the prerequisites NASA already requires of astronauts, the teambuilding and soft-skills training received by mission control personnel will also be required of Moonbase astronauts. This would prepare them to make informed decisions as a team, ensuring that all of the varied expertise between them is used.

Currently, one of NASA's goals for astronaut selection is to find people who aren't shy to do what either they want to do, or what needs to be done. It's important for astronauts to be capable of voicing their opinions, choosing tasks in an efficient way, and making their needs known.

One goal of the crew selection in this report is to show which characteristics I personally satisfy in the selection process:

- Adaptability: As a student and throughout my life I've always been able to learn new skills at a fast rate and enjoy doing so. This would be important for an astronaut, switching between different fields such as geology, piloting, medicine quickly.
- Expressivity: Keeping in tune with my feelings and mental state has been something I've worked at for a long time, and expressing this state to others would be important during the process of adapting to the difficulties of a long-term mission.
- Tasking: I've always been good at staying ahead of tasking difficulties, as I am usually able to prioritize in a way that reduces deadline pressure. This would be more important than convenience on a Moonbase, so astronauts would need to have a good handle on their own capabilities.

A16.0 LUNAR LAUNCH AND LANDING PILOTING

The lunar landings of Apollo 11,12,14, and 15 are compared below on the basis of dust kickup:

- Amount of dust kicked up on landing
 - Apollo 11: The first dust seen flows over the lip of a crater away from the lander. A larger sheet of dust begins to form at around 75 ft.
 - Apollo 12: The first dust seen flows over a flat plain sprawling in front of the lander.
 - Apollo 14: First dust flies over the lip of a crater downrange of the lander. Seemingly a theme of these landings is that dust kicks over craters before the wider sheet of dust forms over the surface.
 - Apollo 15: Larger dust sheet shows up before coming over any craters. Breaks the theme of the last three landings.
- Altitude callout closest to first dust kickup

- Apollo 11: 120 ft
- Apollo 12: 120 ft
- Apollo 14: 110 ft
- Apollo 15: 120 ft
- Discernibility of the boulders and craters near landing site
 - Apollo 11: Boulders are easily discernible throughout the landing site
 - Apollo 12: Boulders seem sparse
 - Apollo 14: Boulders seem sparse
 - Apollo 15: Only one boulder is seen
- Other observations
 - Apollo 11: Faster landing than other Apollos
 - Apollo 12: Longer hover time than on Apollo 11
 - Apollo 14: Landed on a slope
 - Apollo 15: Far more cratered surface

The rate of descent and landing techniques for each of the following vehicles were compared:

- Apollo Lunar Module: Very slow, controlled descent with multiple abort opportunities built into the landing. Most abort chances, highest propellant use option.
- SpaceX: Very fast, “suicide burn” style, less chance for abort options in case of mechanical failures. Least abort chances, least propellant usage option.
- Blue Origin: Pretty fast, some hover time above surface. A good mix between the Apollo and SpaceX options.

In my opinion, the Apollo LM option would be the best as-is, giving the most time margin for troubleshooting and options for abort scenarios. SpaceX and Blue Origin's options could be adapted to add more time for decision making, so the extra potential risk of using these designs could be mitigated. SpaceX's Starship will be used due to its heavy lift capacity.

Me on the Moon!

